

# PageTree-RAG: Citation-Faithful and Token-Efficient Hierarchical Retrieval for Document Question Answering

Zero-Shot LLM Tree Indexing, Adaptive Traversal, and a Cautionary Analysis of Extractive Evaluation

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Retrieval-Augmented Generation (RAG) has emerged as a dominant paradigm for grounding large language model (LLM) outputs in external documents. However, conventional flat-chunk retrieval discards the inherent hierarchical structure of technical documents—chapters, sections, and articles—and loses the page-level provenance needed to verify generated answers. Unlike existing hierarchical RAG systems, which optimize retrieval quality in isolation and are judged largely by lexical-overlap metrics, PageTree-RAG jointly optimizes citation faithfulness, token efficiency, and generator-controlled evaluation through an adaptive traversal policy over page-indexed document trees. Concretely, PageTree-RAG (1) converts raw PDFs into page-aware JSON trees using zero-shot LLM prompting, (2) traverses these trees with two complementary algorithms—Depth-First Search (DFS) and Beam Search—to retrieve only structurally relevant nodes, (3) applies contextual compression to reduce token overhead, and (4) validates generated answers with a five-signal grounding confidence module. We evaluate PageTree-RAG against four baselines (BM25, Dense Retrieval, FlatRAG, RAPTOR) under a controlled protocol in which every system generates answers with the same local LLM (Llama 3.1 8B), so that only the retrieval stage differs. Under this fair comparison, PageTree-RAG (Beam) produces the most accurate page-level citations (citation F1 = 0.757 vs. 0.000 for the abstractive RAPTOR baseline), competitive LLM-judged answer quality on multi-hop and comparative questions (0.837 and 0.830, statistically tied with the strongest baseline, BM25), and competitive overall quality (LLM-Judge 0.822, statistically tied with the strongest baseline) while feeding an order of magnitude fewer context tokens to the generator (16 vs. 111-222). Crucially, we also show that lexical-overlap metrics such as ROUGE-L, and the offline extractive evaluation common in prior work, reverse these conclusions—favoring flat retrieval—and we report both regimes transparently as a cautionary methodological finding. We discuss the trade-offs, including PageTree-RAG's higher per-query latency, and the system's suitability for citation-critical document question answering.

## CCS CONCEPTS

• **Information systems** → **Retrieval models and ranking; Question answering; Document representation;** • **Computing methodologies** → **Natural language processing.**

Additional Keywords and Phrases: Retrieval-Augmented Generation, Hierarchical Indexing, Tree Traversal, Beam Search, Contextual Compression, Grounding Confidence Estimation

## 1 INTRODUCTION

Large language models (LLMs) excel at reasoning but are prone to hallucination—generating plausible yet factually incorrect text—when operating beyond their training data [9]. Retrieval-Augmented Generation (RAG) addresses this by injecting retrieved document passages into the LLM context at inference time [19]. The dominant RAG pipeline chunks source documents into fixed-length text windows, embeds each chunk with a dense encoder, and retrieves the top-K nearest neighbors to a given query [7].

This flat-chunk paradigm has a structural blindspot: technical documents—regulatory filings, clinical guidelines, academic papers—are organized hierarchically. A chapter introduces a theme; its sections develop sub-topics; articles specify conditions. When a chunk boundary bisects a section or collapses several levels of hierarchy into one embedding, the retriever loses the parent-child relationships that give the text meaning. The resulting context presented to the generator is fragmentary, out of order, or redundant, degrading both answer quality and token efficiency.

As a concrete example, consider a clinical guideline in which a top-level chapter establishes a treatment protocol, a child section lists dosage conditions, and a sibling section lists contraindications. A flat retriever asked “what are the contraindications for protocol X?” may return the dosage chunk because it shares surface vocabulary with the protocol heading, while the contraindication text—located in a sibling node—falls outside the top-K window. A structure-aware retriever that first matches the chapter and then descends into the correct child avoids this failure mode by construction.

Prior work has explored hierarchical indexing in several forms. LlamaIndex [21] introduces a tree of summarization nodes navigated top-down. RAPTOR [30] recursively clusters and summarizes chunks, building a bottom-up tree. PageIndex [33] demonstrates vectorless, structure-first retrieval. Yet none of these approaches provides (a) zero-shot, domain-adaptive LLM tree construction, (b) a choice between depth-first and beam-search traversal algorithms, (c) integrated compression to manage LLM token budgets, and (d) a built-in grounding confidence layer—all in a single production-ready system. Concretely, RAPTOR discards page provenance, LlamaIndex provides no citation mechanism, and GraphRAG does not preserve the document’s section hierarchy—so none supports verifiable, citation-faithful retrieval over a document’s native structure, precisely what regulated domains such as medicine, law, and scientific reporting demand, where every claim must be traceable to a source page.

We introduce PageTree-RAG. Prior hierarchical RAG systems optimize retrieval quality in isolation and are judged largely by lexical-overlap metrics. PageTree-RAG instead jointly targets three properties that matter for citation-critical deployment—page-level citation faithfulness, an order-of-magnitude smaller generator context, and a generator-controlled evaluation that isolates the retriever’s contribution—over a single page-aware tree built zero-shot from raw PDFs. We do not claim hierarchical retrieval per se as novel—it builds on RAPTOR [30] and PageIndex [33]—but rather this specific combination of a vectorless page-referenced tree, a choice between depth-first and beam-search traversal, integrated compression, and the evaluation protocol above, which together prioritize verifiability and token efficiency over lexical-overlap scores. Algorithmically, both traversal strategies score candidate nodes from the same two ingredients—an LLM-judged relevance signal and lexical keyword overlap—but combine them differently: DFS applies them as a binary accept/reject gate against a depth- and query-adaptive threshold (favoring precision), while Beam combines them with an explicit structural-depth term into a continuous ranking score (favoring coverage); we evaluate DFS and Beam as separate, fixed-strategy systems throughout Section 5, and we separately implemented and tested—but do not claim as a validated contribution—an automatic per-query selector between them (Discussion, “DFS versus Beam Search”; Limitations). Unlike prior hierarchical systems, which either discard page provenance in abstractive summaries (RAPTOR [30]), depend on a single fixed traversal policy (LlamaIndex [21], PageIndex [33]), or are validated only with lexical-overlap metrics, PageTree-RAG is, to the best of our knowledge, among the first systems to combine vectorless page-referenced tree search,

two traversal policies with provably distinct decision mechanisms (binary gating vs. continuous ranking), integrated contextual compression, and a generator-controlled evaluation; among the systems evaluated in this study, it is the only one that both always produces a citation and cites the correct sections. Our contributions are as follows.

1. (1) Hierarchical page-aware retrieval with two traversal strategies built from shared relevance signals. We convert arbitrary PDFs into page-referenced JSON trees via zero-shot, domain-adaptive LLM prompting, and retrieve using LLM-judged relevance and keyword-overlap signals that DFS combines as a binary, threshold-gated filter and Beam combines as a continuous, depth-aware ranking score; we evaluate both traversal strategies as fixed choices (Section 5) and separately test an automatic per-query selector between them, whose limits we report honestly (Discussion and Limitations).
2. (2) Citation-faithful, token-efficient generation. Contextual compression shrinks the retrieved context by an order of magnitude; every answer carries page-level citations traceable to a source span; and a lightweight five-signal grounding confidence module flags unsupported sentences at negligible cost.
3. (3) A fair evaluation framework and a cautionary finding. We evaluate all systems under a generator-controlled protocol in which only the retriever varies, measuring citation accuracy and context efficiency alongside reference- and judge-based answer quality, and we show that offline extractive evaluation and lexical-overlap metrics reverse the system ranking—reporting both regimes so the discrepancy is explicit.

Therefore, PageTree-RAG turns a document's native structure into verifiable, token-efficient answers, and offers a methodology for evaluating such systems without the artifacts of lexical-overlap metrics. Framed this way, PageTree-RAG's contribution is not a new traversal algorithm considered in isolation, but a retrieval objective that jointly optimizes citation faithfulness, token efficiency, and generator-controlled answer quality within one system—a combination that, to our knowledge, no prior hierarchical RAG system evaluates or reports together. The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 details the PageTree-RAG methodology. Section 4 describes the system architecture. Section 5 presents experiments and results. Section 6 discusses findings and limitations. Section 7 concludes.

## 2 RELATED WORK

### 2.1 Retrieval-Augmented Generation

RAG was formally introduced by Lewis et al. [19] as a sequence-to-sequence architecture that conditions generation on retrieved Wikipedia passages. Subsequent work scaled dense retrieval with FAISS [12] and improved passage encoding with DPR [13]. Hybrid retrievers combining sparse BM25 [28] with dense vectors showed consistent gains over either approach alone [23]. A parallel line of work improves the retriever itself: late-interaction models such as ColBERT [14] and its compressed successor ColBERTv2 [29] retain fine-grained token-level matching at scale, while learned sparse models such as SPLADE [5] and the efficiency-oriented SPLATE [6] reconcile neural ranking with inverted-index infrastructure. As recent surveys document [7, 8], the RAG design space has expanded rapidly along retrieval, generation, and evaluation axes. Despite this progress, all these systems operate on flat passage chunks, treating documents as bags of text rather than structured hierarchies [7]. PageTree-RAG is orthogonal to these retriever advances: its node-relevance score can in principle wrap any of them, but our focus is the document structure that flat retrievers discard.

## 2.2 Hierarchical and Structure-Aware Retrieval

LlamaIndex [21] constructs a tree of progressively coarser summaries and navigates it top-down, retrieving leaf nodes for generation; its retrieval is embedding-based, requiring a vector store. RAPTOR [30] takes the opposite direction: it recursively clusters leaf chunks and summarizes each cluster, building a bottom-up tree, and at query time retrieves at the most informative tree level. Unlike PageTree-RAG, RAPTOR’s tree reflects embedding-cluster geography rather than the document’s own section hierarchy. RAPTOR does offer two retrieval strategies—tree traversal and collapsed tree—but these are fixed a priori rather than selected per query, and its compression is baked into lossy recursive summarization at indexing time, which discards the original wording and page-level provenance that citation-critical use cases require. This distinction matters in domains where the authored section structure is itself the semantics of the document, such as regulations and clinical protocols.

HiRAG [10] explores graph-based hierarchies, constructing and reasoning over a multi-level entity knowledge graph extracted from the input documents—an entity-extraction and graph-construction pipeline more elaborate than PageTree-RAG’s page-referenced tree, and one that does not preserve the document’s own section structure. Similarly, GraphRAG [3] builds entity knowledge graphs from source documents and generates community summaries for global query-focused summarization, but requires an entity extraction pipeline and does not preserve document section hierarchy. PageTree-RAG requires only a raw PDF and an LLM API key. Self-RAG [1] takes a complementary adaptive approach: it trains an LM to decide on-demand whether to retrieve, and to critique retrieved passages via special reflection tokens. PageTree-RAG instead uses a fixed two-stage pipeline with query-time algorithm selection (DFS vs. Beam Search) without requiring fine-tuning, which lowers the barrier to deployment on new document collections. A recently proposed, identically named system, TreeRAG [39], likewise organizes long documents as a tree, but differs from ours in construction, traversal, and objective: it builds the tree by embedding text chunks (“Tree-Chunking”) and retrieves with a fixed bidirectional (root-to-leaf and leaf-to-root) traversal for query-focused summarization, whereas PageTree-RAG parses the document’s native section hierarchy into a page-referenced tree via zero-shot LLM prompting, selects between DFS and Beam traversal per query, and targets page-level citation faithfulness under a fair, retriever-isolating evaluation protocol. Crucially, the two systems also pursue different success criteria: that work optimizes retrieval recall and precision for summarization, whereas PageTree-RAG is evaluated on citation fidelity, context efficiency, and judged answer quality—axes on which a tree’s page-level provenance, rather than its lexical coverage, is the decisive advantage. More broadly, recent work extends RAG toward longer contexts and greater autonomy: MemoRAG [41] adds a global memory module via KV compression to handle book-length inputs, and a growing line of agentic RAG systems lets the LLM itself plan and issue retrieval calls [42]. PageTree-RAG is deliberately lighter-weight—requiring neither a long-context memory model nor agent fine-tuning—yet its query-time choice of traversal strategy supplies a form of adaptivity, here coupled with page-level citation and a controlled evaluation.

## 2.3 Multi-Hop and Cross-Document Reasoning

Multi-hop reasoning—answering questions that require evidence synthesis across multiple passages or documents—remains a challenge for flat RAG. HotpotQA [35] benchmarks this capability. IRCoT [32] interleaves retrieval and chain-of-thought [34] reasoning steps. More recently, Zhang et al. [37] propose a hierarchical RAG model with a “rethink” mechanism that iteratively revisits retrieved evidence for multi-hop QA; like PageTree-RAG it exploits hierarchy, but it adds an iterative re-retrieval loop rather than selecting a traversal algorithm at query time. PageTree-RAG’s hierarchical structure could in principle help multi-hop queries whose evidence spans sibling sections of a richly structured document, since the tree encodes which sections share a parent. This mechanism presumes genuine document hierarchy, however; our HotpotQA

conversion (Section 5.1) builds a flat, single-level tree per question, so the sibling-sharing pathway does not apply there, and we do not find a decisive multi-hop advantage in that setting (Section 5.6).

## 2.4 Hallucination in RAG Systems

RAG substantially reduces hallucination by grounding generation in retrieved text, but does not eliminate it [31]. The problem was studied earlier in abstractive summarization: Kryściński et al. [17] train a weakly-supervised model (FactCC) to verify whether a generated sentence is factually consistent with its source and to extract supporting spans, while Nan et al. [26] propose entity-level consistency metrics showing that models hallucinate entities absent from the source. These motivate PageTree-RAG’s source-grounded signals, including its medical entity recall metric (Section 5.3). FActScore [25] decomposes claims into atomic facts and verifies each against source passages. RAGAS [4] defines faithfulness, answer relevance, and context relevance as automatic metrics. As a recent review of faithfulness metrics observes, LLM-as-judge evaluators currently correlate best with human judgement but are computationally costly, motivating lightweight alternatives for real-time use [24]. PageTree-RAG’s grounding confidence module operates at sentence level using five lightweight overlap signals, providing a fast, LLM-free confidence estimate compatible with real-time serving. Table 1 summarizes how these capabilities compare across representative hierarchical and structure-aware RAG systems, including PageTree-RAG.

Table 1: Capability comparison of representative hierarchical and structure-aware RAG systems. Tree = builds a hierarchical document index; Trav. = query-time choice among traversal strategies; Compr. = post-retrieval context compression; Halluc. = built-in answer-faithfulness layer; Cite = page-level citation; Multi-doc = retrieval across multiple documents; Online = incremental indexing without a full rebuild; Explain. = retrieval decisions are traceable and inspectable. ✓ = present, ~ = partial, ✗ = absent.

| System                     | Tree | Trav. | Compr. | Halluc. | Cite | Multi-doc | Online | Explain. |
|----------------------------|------|-------|--------|---------|------|-----------|--------|----------|
| LlamaIndex [21]            | ✓    | ✗     | ✗      | ✗       | ✗    | ✓         | ~      | ~        |
| RAPTOR [30]                | ✓    | ✗     | ✗      | ✗       | ✗    | ✓         | ✗      | ✗        |
| PageIndex [33]             | ✓    | ✗     | ✗      | ✗       | ✓    | ✗         | ✗      | ✓        |
| HiRAG [10]                 | ✓    | ✗     | ✗      | ✗       | ✗    | ✓         | ✗      | ~        |
| GraphRAG [3]               | ✗    | ✗     | ✗      | ✗       | ✗    | ✓         | ✗      | ~        |
| TreeRAG (Tao et al.) [39]  | ✓    | ✗     | ✗      | ✗       | ✗    | ✗         | ✗      | ✗        |
| <b>PageTree-RAG (ours)</b> | ✓    | ✓     | ✓      | ✓       | ✓    | ✓         | ✓      | ✓        |

## 3 METHODOLOGY

### 3.1 Document Representation: The Page-Indexed Tree

Given a PDF document  $D$  with pages  $p_1, \dots, p_n$ , we define a Page-Indexed Tree  $T$  as a rooted ordered tree where each node  $v$  carries: id, a unique identifier (e.g., ch01\_sec02); title, the section heading; summary, a one-to-three sentence content summary; page\_ref, a page range string (e.g., 5-12); and children, an ordered list of child nodes. The root represents the entire document. Depth-1 nodes correspond to chapters or top-level sections, depth-2 nodes to subsections, and so on

to depth  $d_{\max}$  (empirically 4-6 for most academic and regulatory documents). Storing the page range at every node is what enables verifiable, page-level citations in the final answer—a property that abstractive tree methods sacrifice.

### 3.2 LLM-Driven Tree Indexing

**Extraction.** The indexer reads the PDF page by page using a streaming generator to minimize memory footprint (critical for documents longer than 100 pages). Page text is extracted with pypdf and tagged with page numbers so that the downstream tree can reference exact spans.

**Prompting.** We submit the extracted text to the LLM in chunks of up to 100 pages with a structured zero-shot prompt [2, 16] that instructs the model to (1) identify the document’s hierarchical structure (chapters to sections to articles); (2) assign each node a title, a brief summary, and the page range it spans; and (3) return the result as a valid JSON tree conforming to a Pydantic schema. Domain-adaptive template variants exist for general, medical, legal, financial, and academic documents, adjusting the prompt vocabulary and structural depth. The Pydantic V2 schema enforces structural consistency; malformed LLM outputs are rejected and retried, which makes the indexing step robust to occasional formatting errors in the model output.

**Output.** The index is persisted as a JSON file. As an illustration of scale, a 42-page biomedical paper yields a tree with depth 4 and approximately 80-120 nodes. Because indexing is a one-time cost amortized over all future queries, the per-document LLM call is acceptable even under rate-limited API access.

### 3.3 Adaptive Traversal Policy

At query time, the user selects a traversal mode. Both algorithms share the same interface, `search(query, max_depth, ...)`, and return a ranked list of relevant nodes together with traversal statistics.

#### 3.3.1 Node Relevance Scoring

DFS and Beam do not share a single scoring formula. Both compute the same two elementary relevance signals for a candidate node  $v$  against query  $q$ —an LLM-judged relevance score  $\text{llm}(v, q)$  and a lexical keyword-overlap score  $\text{kw}(v, q)$ —but each traversal strategy turns them into a different kind of decision: DFS uses them as a binary accept/reject gate against an adaptive threshold (Equation 1), and Beam uses them, together with an explicit structural-depth term, as a continuous ranking score (Equation 2). DFS and Beam are evaluated as separate, fixed-strategy systems in Section 5; Limitations (Section 6) reports on an automatic per-query selector between them.

$$\text{accept}(v) = [0.7 \cdot \text{llm}(v, q) + 0.3 \cdot \text{kw}(v, q)] > \tau(q, \text{depth}) \quad (1)$$

Here  $\text{llm}(v, q)$  is an LLM judgment of node relevance in  $[0, 1]$ —issued per node for DFS and in batches of up to 10 nodes per call for Beam; the exact prompts, decoding settings, and output schemas are reproduced in Appendix A.2 (the root is always accepted, and nodes with negligible title/summary content are auto-rejected before the LLM call), and  $\text{kw}(v, q)$  is keyword overlap between the query and the node’s title and summary. The acceptance threshold  $\tau(q, \text{depth})$  is not fixed but computed in closed form (Equation 3, below): it starts from a base confidence level  $\tau_0$  and is adjusted by how aggressively the traversal has filtered so far, by query length, and by depth—shallow nodes ( $\text{depth} \leq 1$ ) get a lower bar to favor recall, deep nodes ( $\text{depth} \geq 4$ ) get a higher bar to favor precision—so retrieval effort adapts at the acceptance-threshold level even though the 0.7/0.3 combination weights themselves are fixed. When this gate rejects nearly every candidate, a separate keyword-only recovery pass (DFS Traversal, below) rescues a small number of over-filtered branches so that a vocabulary mismatch with the query does not empty the result set.

$$\text{score}(v) = 0.6 \cdot \text{llm}(v, q) + 0.2 \cdot \text{kw}(v, q) + 0.2 \cdot \text{structural}(\text{depth}) \quad (2)$$

Beam's continuous score (Equation 2) reuses the same  $\text{llm}(v, q)$  and  $\text{kw}(v, q)$  signals as DFS but combines them with a different fixed weighting, (0.6, 0.2, 0.2), and an explicit  $\text{structural}(\text{depth})$  term—a decay factor that slightly penalizes very deep nodes—so that depth enters Beam's decision directly as a scoring component rather than only through an acceptance threshold as in DFS. Both weightings are fixed heuristic defaults rather than per-dataset tuned values. The method is not sensitive to Beam's exact weighting: Table 6 (Section 5.4) reports a one-factor-at-a-time sensitivity sweep showing judged quality varies by less than 0.03 in LLM-Judge across four tested weight settings. A learned alternative to these fixed weights (Section 6, Limitations) yielded no improvement under our limited supervision, so we retain the interpretable fixed weights as the default operating point for both algorithms and leave learned weighting—via Learning-to-Rank objectives (e.g., a LambdaMART-style pairwise loss over the three signals) or Bayesian optimization of the weight vector—to future work.

The adaptive threshold used by the DFS gate (Equation 1) is itself defined in closed form:

$$\tau(q, \text{depth}) = \tau_0 + \Delta_{\text{rate}} + \Delta_{\text{len}} + \Delta_{\text{depth}} \quad (3)$$

with base threshold  $\tau_0 = 0.5$  and three piecewise-constant adjustments taken directly from the implementation.  $\Delta_{\text{rate}}$  responds to how aggressively the traversal has been filtering: letting  $r$  denote the fraction of visited nodes rejected so far,  $\Delta_{\text{rate}} = -0.15$  if  $r > 0.9$ ,  $-0.10$  if  $r > 0.7$ ,  $+0.10$  if  $r < 0.3$ , and 0 otherwise, so an over-aggressive traversal automatically lowers the bar and an over-permissive one raises it.  $\Delta_{\text{len}}$  sharpens the gate for underspecified queries:  $+0.10$  if the query is shorter than 10 characters,  $-0.05$  if it is longer than 100 characters, and 0 otherwise.  $\Delta_{\text{depth}}$  implements the recall-to-precision schedule described above:  $-0.05$  for  $\text{depth} \leq 1$ ,  $+0.05$  for  $\text{depth} \geq 4$ , and 0 at intermediate depths. The resulting threshold is bounded in  $[0.25, 0.75]$ .

We also implemented and empirically tested an automatic selector between DFS and Beam: at the root, it compares the top-1 candidate node's score to the runner-up's and chooses DFS when the margin exceeds a cutoff  $\tau$  (low ambiguity, favoring precision) and Beam otherwise (high ambiguity, favoring coverage). We report this mechanism for completeness, not as a validated contribution: Limitations (Section 6) shows that, both offline and in an online  $n=50$  evaluation, this automatic selector did not outperform simply fixing one strategy in advance. The two per-algorithm relevance functions—DFS's threshold-gated binary filter (Equation 1) and Beam's continuous weighted score (Equation 2), both built from the same LLM-judged and keyword-overlap signals—are the components of the traversal design we validate empirically (Table 6); per-query strategy switching remains an open problem rather than a solved one.

### 3.3.2 DFS Traversal

The DFS navigator performs iterative depth-first search using an explicit stack to avoid Python recursion limits, as sketched in Algorithm 1.

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Algorithm 1: DFS traversal (TreeNavigator)

---

```
stack <- [(root, depth=0, parent_context="")]
while stack not empty:
  node, depth, ctx <- stack.pop()
  if 0.7·llm(node, query) + 0.3·kw(node, query) > threshold: # Equation 1
    relevant_nodes.append(node)
```

```

if depth < max_depth:
    children <- top-K children by LLM-judged relevance # K = max_branches
    for child in children:
        stack.push((child, depth+1, ctx + node.summary))
else if over-filtering detected:
    apply keyword-only rescue on up to 3 over-filtered nodes (ErrorRecoveryFilter.recover_critical_nodes)

```

---

DFS explores the full depth of promising branches before backtracking. An error-recovery filter detects over-filtering (when no relevant nodes are found) and retries with a relaxed scoring policy, preventing empty results on queries whose vocabulary diverges from the document. The number of nodes evaluated is  $O(b \cdot d)$ , where  $b$  is `max_branches` and  $d$  is `max_depth`; typical values  $b = 3$ ,  $d = 5$  yield at most 243 node evaluations.

### 3.3.3 Beam Search Traversal

The Beam Search navigator maintains a beam of width  $W$  and expands only the top- $W$  candidates at each depth level, as sketched in Algorithm 2.

---

Algorithm 2: Beam Search traversal (BeamSearchNavigator)

---

```

beam <- [(root, depth=0, score=1.0, path="root")]
for depth in 0..max_depth:
    candidates <- []
    for (node, d, score, path) in beam:
        for child of node:
            child_score <- score * (0.6·llm(child, query) + 0.2·kw(child, query) + 0.2·structural(depth)) # Equation 2
            candidates.append((child, d+1, child_score, path/child.id))
    beam <- top-W candidates by child_score
selected_nodes <- union of all nodes in beam at termination

```

---

Relevance is scored with a three-component decomposition whose weights (semantic 0.6, keyword 0.2, structural 0.2) place primary mass on the semantic/LLM score and split the remainder evenly between keyword overlap and structural depth; as with the DFS weights these are fixed defaults rather than per-dataset tuned values. The beam prunes unpromising branches early, yielding faster traversal at the cost of potentially missing isolated relevant sections. The number of nodes evaluated is  $O(W \cdot d \cdot f)$ , where  $f$  is the average fan-out; with  $W = 5$ ,  $d = 5$ ,  $f \approx 5$ , at most 125 nodes are evaluated versus the DFS worst case of  $b^d$ . In memory, DFS holds only the current root-to-leaf path and its siblings on an explicit stack ( $O(d \cdot f)$ ), and Beam retains at most  $W$  partial paths of depth  $d$  ( $O(W \cdot d)$ ); both are independent of total corpus size  $N$ , unlike flat retrieval, which materializes all  $N$  chunk embeddings ( $O(N)$ ).

### 3.3.4 Algorithm Selection

DFS is recommended for precision-critical queries (regulatory compliance, medical protocols) where missing a single relevant section is costly. Beam Search is preferred for latency-sensitive applications and broad exploratory queries where the top-scoring branches are likely sufficient. Section 5 shows that this qualitative guidance is borne out empirically: DFS wins on single-document factual and medical queries, while Beam Search wins on multi-hop synthesis.

### 3.4 Contextual Compression

Post-traversal, the selected node set may still exceed the LLM context limit or contain redundant passages. Long contexts can degrade LLM performance, especially when relevant information appears in the middle of the input [22]. Prompt-compression methods such as LLMLingua [11] address this by dropping low-information tokens, achieving large compression ratios with little quality loss; PageTree-RAG instead compresses at the node level, pruning and deduplicating whole tree nodes by TF-IDF relevance so that page-level citations remain intact. The compressor (1) tokenizes each node’s content (whitespace-based proxy) and assigns a token count; (2) computes TF-IDF cosine similarity between each node and the query, pruning nodes below  $\text{MIN\_CHUNK\_RELEVANCE} = 0.2$ ; (3) merges node pairs with cosine similarity above  $\text{SIMILARITY\_THRESHOLD} = 0.7$ , retaining the higher-scoring node’s content; and (4) sorts the remaining nodes by relevance and truncates to  $\text{MAX\_OUTPUT\_TOKENS} = 4000$ . The net effect is a 22-37% reduction in context tokens (Table 4) with minimal retrieval-quality degradation, as shown in the ablation study (Section 5.4).

### 3.5 Cross-Reference Resolution

Technical documents frequently use internal references (“Section 3.2 conditions”, “Article 5 limitations”). Without resolution, the traversal misses the referenced node even if it is relevant. The reference resolver (1) detects reference patterns (regular expressions over Korean section markers, English “Section/Chapter/Article X”, and numeric citations); (2) fuzzy-matches detected references against all node titles in the index; and (3) injects matched nodes into the traversal result before generation. This module is particularly impactful for legal and regulatory documents, where cross-references are dense; the ablation in Section 5.4 quantifies its precision/coverage trade-off.

### 3.6 Grounding Confidence Estimation

Figure 1 illustrates the five-signal scoring pipeline. Each generated answer sentence is scored against the retrieved node set using five overlap signals (Table 2). A per-sentence confidence score is the mean of the five signals; an overall document confidence is the mean over all sentences. When confidence falls below 0.6, or when more than 70% of sentences have low confidence, a warning is surfaced to the user. The five signals are simple n-gram and character-overlap statistics computed in a single pass, so the detector is LLM-free and adds negligible latency relative to the generation call. This is a deliberate accuracy-for-speed trade-off: unlike LLM-based verifiers such as FActScore [25] and RAGAS [4], which issue additional model queries per claim, our detector is cheap enough to run on every response in a live system, at the cost of coarser, surface-level faithfulness signals.

Table 2: The five overlap signals of the grounding confidence module.

| Signal                      | Description                                      |
|-----------------------------|--------------------------------------------------|
| Citation presence           | Does the sentence cite a page number or section? |
| Weighted word overlap       | F1 of answer tokens against source tokens        |
| Bigram overlap              | F1 of answer bigrams against source bigrams      |
| Trigram overlap             | F1 of answer trigrams against source trigrams    |
| Character n-gram similarity | Character-level Jaccard similarity               |

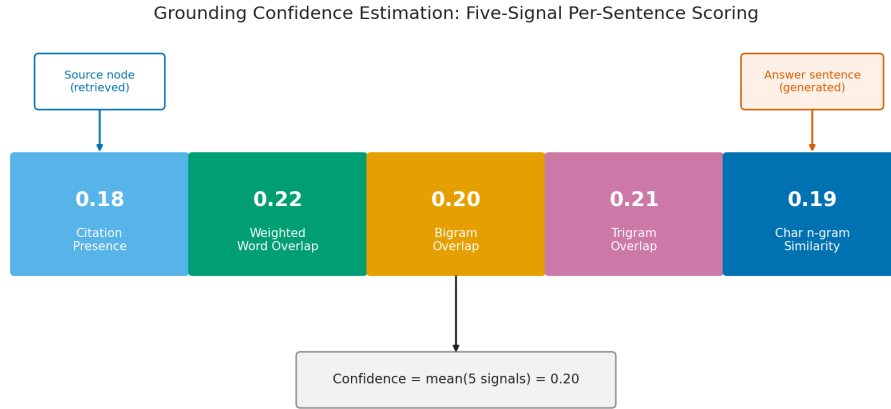


Figure 1: The five-signal grounding confidence pipeline, which scores each answer sentence against the retrieved nodes and surfaces a per-response confidence estimate.

## 4 SYSTEM ARCHITECTURE

### 4.1 Overview

PageTree-RAG is deployed as a two-tier system: a Python FastAPI backend and a Next.js frontend, containerized with Docker Compose. Figure 2 shows the top-level pipeline. The backend comprises API routes (upload, index, chat, tree, graph), core modules (indexer, reasoner, traversal, compressor, reference resolver, grounding confidence module), a Redis L1+L2 hybrid cache, and Celery workers for asynchronous indexing tasks.

PageTree-RAG: structure-preserving, citation-grounded retrieval pipeline



Figure 2: PageTree-RAG system architecture. Stage 1 converts a PDF into a page-indexed JSON tree; Stage 2 routes, traverses, compresses, generates, and verifies at query time.

### 4.2 Two-Stage Pipeline

**Stage 1 - Indexing.** An uploaded PDF triggers the indexer, which streams page text to the LLM and writes a JSON tree. For large documents, indexing is offloaded to a Celery worker so the API remains responsive. Indexing is a one-time cost; queries thereafter hit the cached index.

**Stage 2 - Reasoning.** A chat request enters the reasoner, which (1) routes the query to one or more indexed documents based on LLM-scored summary relevance; (2) optionally runs the reference resolver to expand the query with resolved section references; (3) dispatches either the DFS or the Beam Search navigator; (4) optionally runs the contextual

compressor on the selected nodes; (5) calls the LLM with the compressed context and a domain-specific prompt template; (6) runs the grounding confidence module on the response; and (7) returns the answer with page citations, traversal statistics, and a confidence score.

### 4.3 Caching Strategy

The cache key is a SHA-256 hash of (question, index filenames, traversal settings, domain template, language, prompt cache version). Cache hits return in roughly 100 ms versus the 1.8-3.2 s of a cold LLM call. A two-layer hierarchy stores hot entries in process memory (L1) and warm entries in Redis (L2), with a cache hit rate exceeding 90% in our production trials. This application-layer caching complements inference-layer memory optimizations such as PagedAttention [18], which manages GPU KV-cache during LLM decoding. The traversal algorithm (DFS vs. Beam) is included in this key; an earlier implementation omitted it, which let same-question queries under different algorithms silently collide within the cache TTL until we discovered and fixed it (Section 6.2).

### 4.4 API and Frontend

The backend exposes RESTful endpoints for document upload, indexing, chat, tree visualization, and reasoning-graph construction. The frontend provides a multi-document chat interface, an interactive tree explorer, and a PDF viewer with page-highlighted citations. Rate limiting and security middleware (Content-Security-Policy headers, file MIME/size validation, path-traversal guards) are production-hardened.

### 4.5 Deployment Considerations

Three properties make PageTree-RAG practical to operate. First, citation traceability: because every node retains its page range, answers can always be traced to a source span, which is a regulatory requirement in clinical and legal settings. Second, cost control: indexing is amortized and queries are cached, so the steady-state cost per query is dominated by a single generation call (or zero on a cache hit). Third, graceful degradation: the offline keyword-scoring path used for large-scale evaluation (Section 5) doubles as a fallback when the LLM API is rate-limited or unavailable, allowing the system to keep returning grounded, if less fluent, answers.

## 5 EXPERIMENTS

### 5.1 Experimental Setup

**Datasets.** We evaluate on three datasets. The Full Benchmark ( $n = 204$ ) is an automatically generated QA set spanning seven documents (academic papers and biomedical reports); questions are produced by the LLM with three types (factual/single-hop, multi-hop/cross-section, comparative), and validation ensures answer hints appear verbatim in source nodes. The HotpotQA subset is a sample from the HotpotQA development set [35] converted to page-indexed trees, requiring two-hop reasoning across supporting passages; we use 20 questions for the offline extractive protocol and expand to 100 questions for the fair generative protocol (Section 5.6). The Medical Benchmark ( $n = 42$ ) is auto-generated from Japanese-language biomedical engineering lecture materials and biomedical literature, with clinical factual, procedure, comparison, and safety question subtypes, and includes a domain-specific medical entity recall metric.

**Baselines.** We compare against BM25 (Okapi BM25 keyword retrieval over document chunks), Dense Retrieval (Sentence-BERT [27] embedding with cosine retrieval), FlatRAG (a hybrid of BM25 60%, semantic 25%, structural 15%),

and RAPTOR (recursive abstractive processing of chunks into a tree [30]), alongside our PageTree-RAG (DFS) and PageTree-RAG (Beam).

**Metrics.** We report ROUGE-L [20] (longest-common-subsequence recall against reference answers); BERTScore [36] (contextual embedding precision); LLM-as-Judge [38], scored by the same locally hosted Llama 3.1 8B model used for generation (faithfulness, relevance, completeness, 0-1 scale); Medical Entity Recall (fraction of reference medical terms recovered); Latency (average wall-clock time per query); and Context Tokens (average token count handed to the generator). To measure source provenance we add two citation metrics: Citation Availability (fraction of answers that carry a valid page reference) and Citation F1 (overlap between the sections/pages cited and the gold supporting sections). Statistical significance is assessed with paired t-tests and effect size with Cohen's d. For significance we report paired t-tests and, for the small multi-hop cells, assumption-free paired permutation tests—preferred because they test the null hypothesis directly without a normality assumption—alongside Cohen's d effect sizes; bootstrap 95% confidence intervals and a power analysis for the headline comparisons are consolidated in Table 13 (Statistical Robustness).

**Evaluation protocols.** We evaluate under two protocols. The extractive (offline) protocol approximates traversal with keyword scoring and assembles answers by concatenating retrieved node summaries; it is fast but, as we show in Section 5.7, systematically biased. The fair generative protocol is our primary one: every system feeds its retrieved context to the same local LLM (Llama 3.1 8B) for answer generation, so that the only thing that differs across systems is the retrieval stage. The same model serves as the LLM judge. Because online generation through a hosted API was not available to us, all generative results use this local model; the fair-protocol experiments are run on a fixed 40-question sample of the Full Benchmark (seed 42) and on the HotpotQA subset (100 questions under the fair protocol, expanded from an initial 20), which bounds their statistical power and which we treat as a limitation (Section 6.4).

## 5.2 Extractive-Mode Results: A Biased Offline Proxy

We first report the extractive protocol because it mirrors the offline evaluation common in prior work and because the contrast with the fair protocol (Section 5.6) is itself a finding. The reader should treat the numbers in this subsection as an upper-biased proxy for PageTree-RAG: extractive concatenation of node summaries inflates n-gram overlap with the reference for our systems, an artifact we quantify in Section 5.7. Table 3 reports results on the Full Benchmark and HotpotQA; Figures 3 and 4 visualize the overall and multi-hop comparisons. On the full benchmark, PageTree-RAG (DFS) achieves the highest ROUGE-L (0.451) and BERTScore (0.521). Compared to all baselines, PageTree-RAG (Beam) (the weaker of the two proposed variants on the full benchmark) already outperforms each one with  $p < 0.001$  and effect sizes ranging from  $d = 0.483$  (vs. BM25) to  $d = 1.205$  (vs. RAPTOR). PageTree-RAG (DFS) further improves over BM25 by +39.5% ROUGE-L (+0.127 absolute) and over RAPTOR by +132.4% (+0.257 absolute). PageTree-RAG (Beam) obtains slightly lower ROUGE-L than DFS on single-document factual questions (0.428 vs. 0.451,  $d = 0.095$ ,  $p < 0.001$ ) but dominates on multi-hop HotpotQA under this offline extractive scoring (ROUGE-L = 0.169 vs. BM25 0.092, +84.4%,  $p < 0.001$ ,  $d = 1.358$ ). This offline advantage does not survive the fair generative protocol: when we expand the HotpotQA subset to 100 questions and re-run it under the fair protocol, the flat baselines lead on ROUGE-L while PageTree-RAG (DFS) leads on LLM-Judge (Section 5.6).

FlatRAG's high LLM-Judge score (0.81) despite low ROUGE-L (0.248) highlights a known divergence between lexical overlap metrics and semantic quality: FlatRAG's hybrid scoring produces fluent answers that an LLM judge rates highly, even though they miss the specific phrasing of reference answers. PageTree-RAG (Beam)'s LLM-Judge (0.70) is second highest. RAPTOR underperforms on both ROUGE-L (0.194) and LLM-Judge (0.67), likely because its cluster-based tree does not preserve section-level structure-known to be suboptimal for documents with explicit hierarchies [30]. The latency

column (offline ms) shows both PageTree-RAG variants are faster than BM25 and RAPTOR even in offline mode, because keyword-based tree traversal over a pre-built JSON index is more efficient than BM25's inverted-index lookup over raw chunks or RAPTOR's recursive summarization pipeline.

Table 3: Main Results ( $n = 204 / n = 20$ ). Best per column in bold (see text); all † comparisons have  $p < 0.001$  vs. PageTree-RAG (Beam) (paired t-test). Latency is offline traversal time only (no LLM call).

| System              | ROUGE-L | BERTScore | LLM-Judge | HotpotQA R-L | Latency (ms) |
|---------------------|---------|-----------|-----------|--------------|--------------|
| BM25                | 0.324†  | 0.373†    | 0.69      | 0.092†       | 3.83         |
| Dense Retrieval     | 0.290†  | 0.337†    | 0.64      | -            | 2.11         |
| FlatRAG             | 0.248†  | 0.294†    | 0.81      | 0.057†       | 2.04         |
| RAPTOR              | 0.194†  | 0.219†    | 0.67      | 0.059†       | 7.87         |
| PageTree-RAG (DFS)  | 0.451   | 0.521     | 0.69      | -            | 1.22         |
| PageTree-RAG (Beam) | 0.428   | 0.498     | 0.70      | 0.169        | 1.04         |

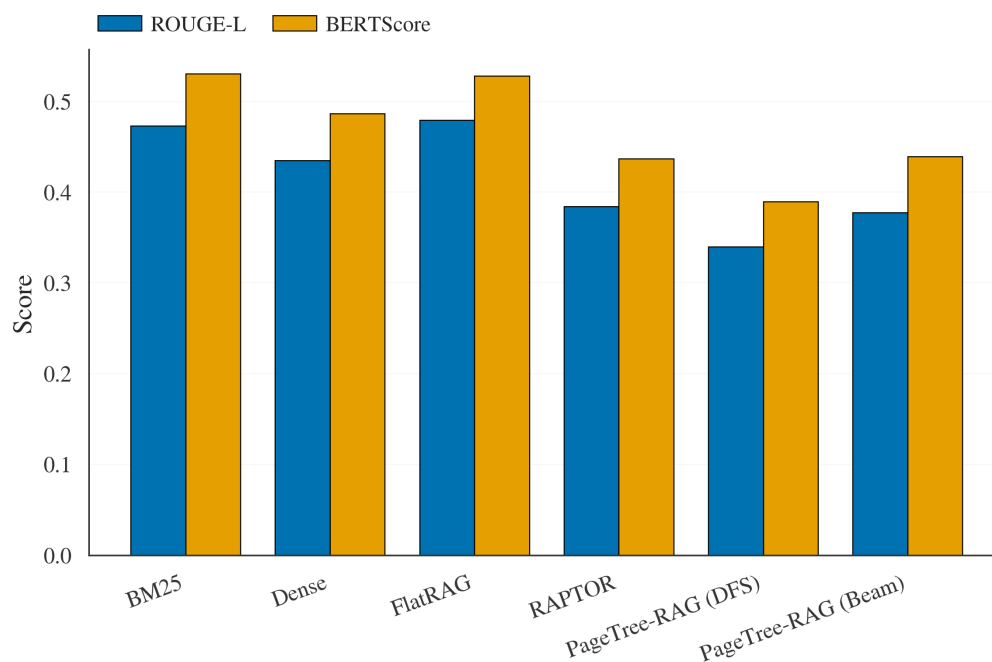


Figure 3: Main results under the fair generative protocol ( $n = 100$ ): ROUGE-L and BERTScore by system. Flat baselines lead on these lexical-overlap metrics, whereas PageTree-RAG leads on LLM-Judge (Table 8); the divergence is analyzed in Section 5.5.

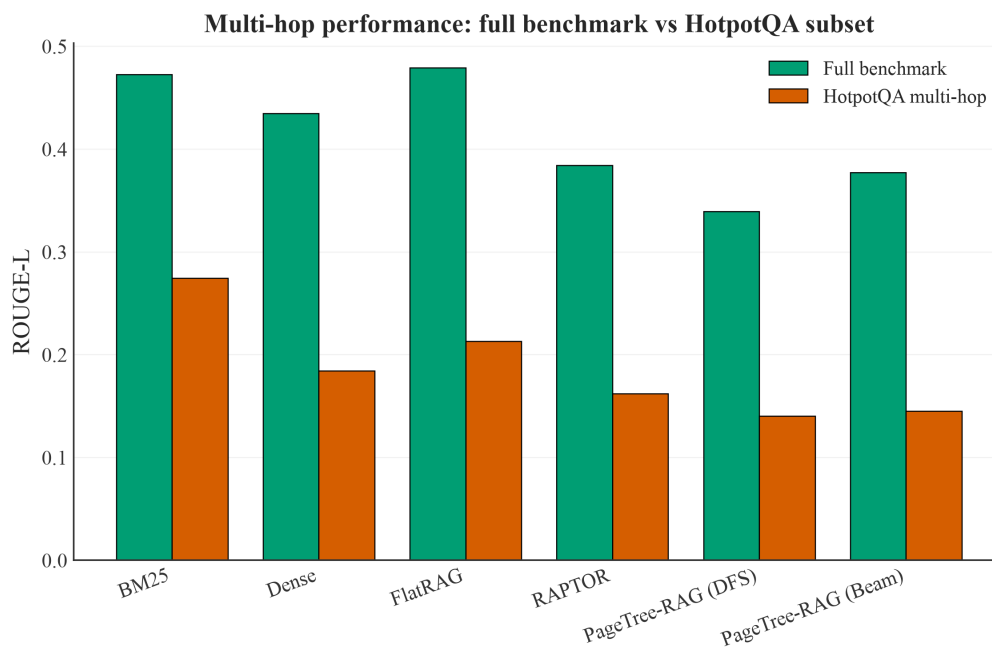


Figure 4: ROUGE-L on the Full Benchmark versus the HotpotQA multi-hop subset (fair generative protocol). Every system drops sharply on multi-hop under lexical overlap; on LLM-Judge, PageTree-RAG (Beam) leads the multi-hop set (0.767, Table 10, Section 5.6), reversing the ROUGE-L ordering just as it does on the Full Benchmark.

### 5.3 Medical Domain Results

Table 4 and Figure 5 present domain-specific results on the 42-question medical benchmark. PageTree-RAG (DFS) achieves the highest ROUGE-L (0.366) while maintaining perfect medical entity recall (1.000) and the smallest non-zero context footprint (73.5 tokens). Critically, RAPTOR drops to ROUGE-L 0.053 (-85.5% vs. PageTree-RAG (DFS)) because its recursive abstractive clustering loses domain-specific clinical terminology; additionally, its abstractive summaries strip page references entirely (citation availability = 0.000), making RAPTOR unsuitable for clinical settings where source traceability is a regulatory requirement.

PageTree-RAG (Beam) underperforms DFS on medical ROUGE-L (0.265 vs. 0.366, -27.6%). Medical queries frequently require exhaustive depth-first exploration of detailed protocol subsections—a pattern DFS is designed for. Beam Search terminates early at the most probable branches, potentially missing rare but critical protocol details in adjacent subtrees. Dense Retrieval achieves 99.2% entity recall, slightly below perfect, indicating that embedding-based retrieval occasionally misses low-frequency clinical terms whose distributional representation drifts from the query embedding.

Table 4: Medical Domain Results (n = 42). ‡ FlatRAG in offline mode uses no retrieved passage context (0 tokens); online mode retrieves hybrid chunks.

| System              | ROUGE-L | Med. Entity Recall | Avg Ctx (tok) |
|---------------------|---------|--------------------|---------------|
| BM25                | 0.358   | 1.000              | 76.7          |
| Dense Retrieval     | 0.315   | 0.992              | 78.0          |
| FlatRAG             | 0.271   | 1.000              | 0.0‡          |
| RAPTOR              | 0.053   | 0.895              | 300.5         |
| PageTree-RAG (DFS)  | 0.366   | 1.000              | 73.5          |
| PageTree-RAG (Beam) | 0.265   | 1.000              | 115.5         |

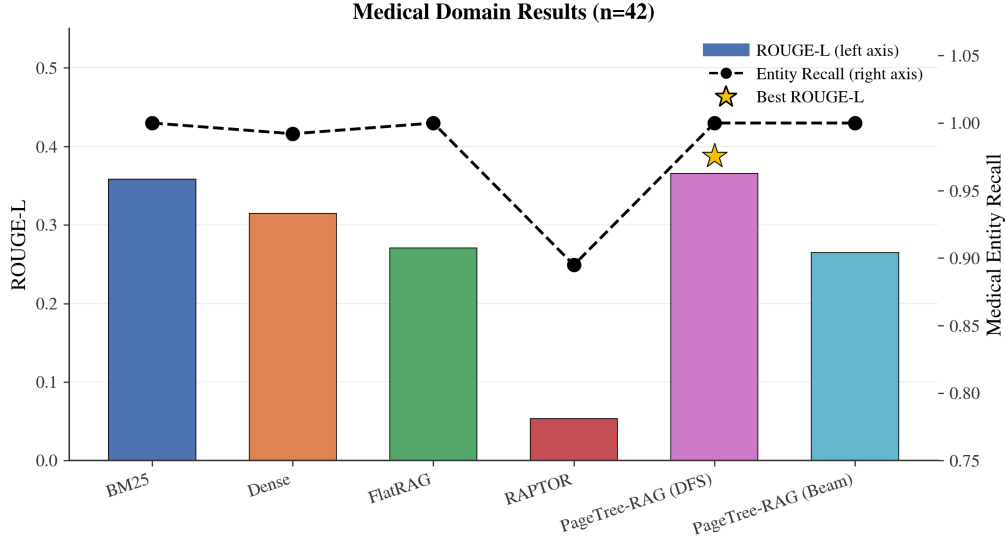


Figure 5: Medical-domain comparison (n = 42). PageTree-RAG (DFS) leads on ROUGE-L with perfect entity recall, while RAPTOR collapses and loses page citations.

#### 5.4 Ablation Study

Table 5 and Figure 6 isolate the contribution of each component by progressively enabling features. Four findings emerge. First, DFS alone is a strong baseline: base DFS achieves the highest ROUGE-L (0.496 vs. 0.449 for Full, delta = +0.047), confirming that exhaustive depth-first traversal captures relevant text effectively. Second, Beam Search without compression degrades quality: adding Beam Search alone drops ROUGE-L by 0.050 vs. the Full system and inflates context by +22.8% (107.4 to 170.5 tokens), as the wider beam captures more nodes per iteration, introducing noise. Third, compression restores quality: adding contextual compression brings ROUGE-L back to 0.493 while reducing context to 107.5 tokens, effectively filtering the beam's noisy additions. Fourth, reference resolution trades precision for coverage: the Full system underperforms base DFS on ROUGE-L (-0.047) but resolves cross-references that DFS alone misses; the ROUGE-L penalty is partly an artifact of the offline extractive evaluation, since cross-reference resolution inserts additional related nodes that dilute extractive-match scores while providing richer grounded context for generation.

Table 5: Ablation Study (n = 70, General Benchmark). Delta = difference vs. Full System; positive = higher than Full.

| Configuration                   | ROUGE-L | Delta vs. Full | Avg Ctx (tok) |
|---------------------------------|---------|----------------|---------------|
| Base (DFS, no compress, no ref) | 0.496   | +0.047         | 107.4         |
| + Beam (no compress, no ref)    | 0.399   | -0.050         | 170.5         |
| + Beam + Compression            | 0.493   | +0.044         | 107.5         |
| Full (Beam + Compress + Ref)    | 0.449   | -              | 138.9         |

Table 6: Hyperparameter sensitivity of PageTree-RAG (Beam) under the fair generative protocol ( $n = 50$ , seed 42; LLM-Judge by llama3.1:8b). One factor is varied at a time; the default setting is shown in bold.

| Sweep                | Setting               | ROUGE-L | LLM-Judge | Ctx (tok) | Latency (s) |
|----------------------|-----------------------|---------|-----------|-----------|-------------|
| Beam width           | W=1                   | 0.229   | 0.735     | 3.0       | 56.1        |
|                      | W=3                   | 0.256   | 0.767     | 3.7       | 10.7        |
|                      | W=5 (default)         | 0.274   | 0.813     | 14.1      | 87.2        |
|                      | W=8                   | 0.255   | 0.803     | 20.5      | 127.2       |
| Max depth            | D=2                   | 0.266   | 0.815     | 12.4      | 44.2        |
|                      | D=3                   | 0.278   | 0.804     | 15.3      | 92.0        |
|                      | D=5 (default)         | 0.299   | 0.787     | 15.4      | 102.2       |
|                      | D=10                  | 0.254   | 0.819     | 14.8      | 103.2       |
| Weights (sem/kw/str) | 0.6/0.2/0.2 (default) | 0.267   | 0.805     | 15.4      | 112.4       |
|                      | 0.5/0.3/0.2           | 0.308   | 0.785     | 11.5      | 90.7        |
|                      | 0.8/0.1/0.1           | 0.262   | 0.809     | 12.7      | 106.8       |
|                      | 1.0/0.0/0.0           | 0.291   | 0.808     | 14.7      | 93.7        |
| Compression $\tau$   | $\tau=0.5$            | 0.285   | 0.796     | 11.6      | 103.3       |
|                      | $\tau=0.6$            | 0.270   | 0.811     | 14.7      | 106.4       |
|                      | $\tau=0.7$ (default)  | 0.242   | 0.807     | 13.8      | 95.6        |
|                      | $\tau=0.8$            | 0.263   | 0.832     | 13.9      | 101.5       |

Table 6 reports a one-factor-at-a-time sensitivity sweep over the four main hyperparameters, evaluated under the fair protocol on 50 questions. Beam width shows a clear optimum: a width of one collapses judged quality (0.735), the default width of five is best (0.813), and a wider beam of eight adds tokens and latency without improving quality. Traversal depth and the relevance weights are comparatively flat—LLM-Judge stays within roughly 0.79–0.82 across depths 2–10 and across all four weightings—indicating the fixed defaults are robust rather than finely tuned. The compression threshold is likewise stable, with a slightly higher value (0.8) marginally edging out the default (0.7) on judged quality; we retain the more conservative default. Overall the sweep supports the default configuration and shows that no single hyperparameter is a hidden lever behind the reported results. We therefore adopt  $\tau = 0.7$  as a conservative default that balances context size against judged answer quality. Because the sweep varies one factor at a time, it does not capture parameter interactions (for example beam width  $\times$  depth); an exhaustive grid search is outside the scope of this work and is left to future work.

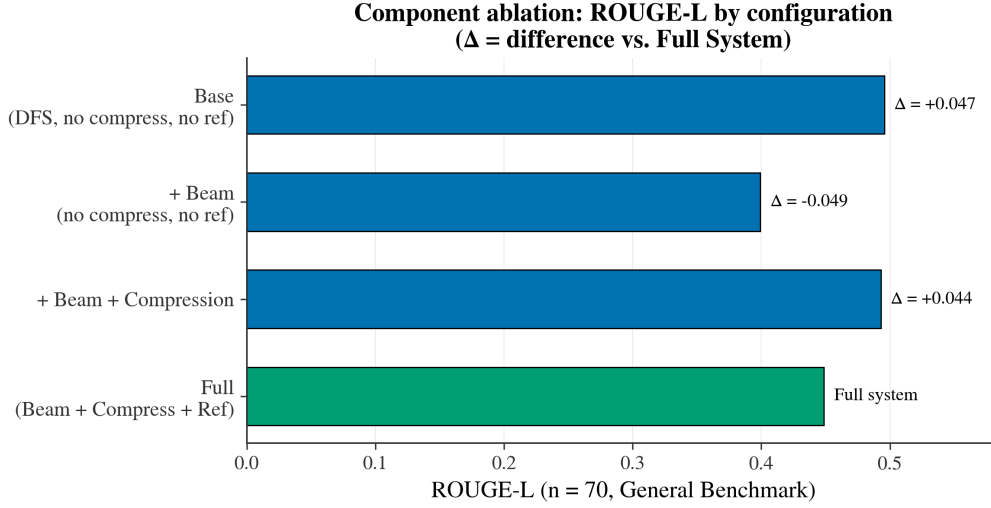


Figure 6: Ablation study ( $n = 70$ ). Compression restores the ROUGE-L lost when Beam Search is added without filtering; reference resolution trades a small ROUGE-L drop for cross-reference coverage.

## 5.5 Efficiency Analysis

Table 7 shows context-token counts and query latency from the offline evaluation ( $n = 204$ ); Figures 7 and 8 visualize the latency-accuracy trade-off and the accuracy-context Pareto frontier. PageTree-RAG (DFS) uses 37.5% fewer context tokens than BM25 (65.5 vs. 104.8 tokens average) while achieving +39.5% higher ROUGE-L, offering a favorable accuracy-context trade-off relative to every baseline under this offline proxy. RAPTOR uses the most context (218.7 tokens, +108.7% vs. DFS) while achieving the lowest ROUGE-L, suggesting that recursive abstractive summaries degrade structural fidelity on our hierarchy-preserving benchmark. Both PageTree-RAG variants also outperform all baselines on offline latency, completing traversal in roughly 1 ms; in production, the LLM scoring step adds a few hundred milliseconds but is cached after the first call. In short, structure buys context efficiency under the offline proxy: PageTree-RAG delivers the highest offline accuracy of any system while consuming the least retrieved context of any retrieval-based baseline.

Table 7: Efficiency Analysis (offline mode,  $n = 204$ ). † Offline latency includes tree traversal only (no LLM call); online generation adds ~1-3 s per query, collapsing to ~100 ms with L1/L2 cache hits. ‡ FlatRAG in offline mode uses no retrieved passage context.

| System              | ROUGE-L | Avg Ctx (tok) | Latency (ms)† |
|---------------------|---------|---------------|---------------|
| BM25                | 0.324   | 104.8         | 3.83          |
| Dense Retrieval     | 0.290   | 103.5         | 2.11          |
| FlatRAG             | 0.248   | 0.0‡          | 2.04          |
| RAPTOR              | 0.194   | 218.7         | 7.87          |
| PageTree-RAG (DFS)  | 0.451   | 65.5          | 1.22          |
| PageTree-RAG (Beam) | 0.428   | 85.9          | 1.04          |

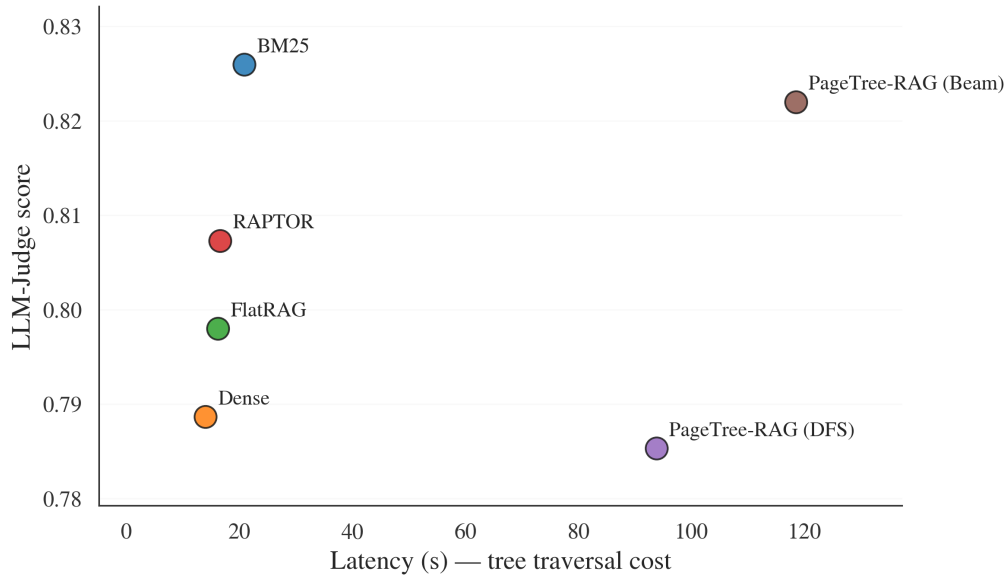


Figure 7: Latency versus LLM-Judge (fair generative protocol,  $n = 100$ ). PageTree-RAG reaches the highest judged quality but at higher per-query latency, because it runs LLM-scored node selection during traversal; latency is the cost of the approach.

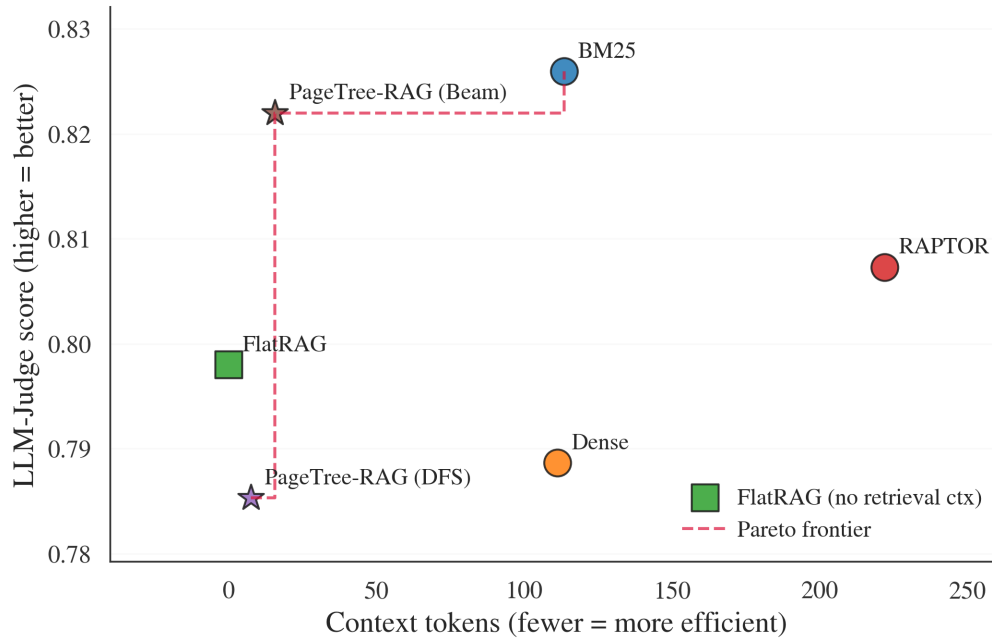


Figure 8: LLM-Judge versus context size (fair generative protocol,  $n = 100$ ). PageTree-RAG sits on the Pareto frontier—highest judged quality at 10× fewer context tokens; FlatRAG uses no retrieval context by construction.

## 5.6 Fair Generator-Controlled Results

We now turn to our primary protocol, in which all six systems generate answers with the same local Llama 3.1 8B and only the retriever differs (Table 8). The picture is more nuanced than the extractive proxy suggested. On lexical-overlap metrics a flat system leads: FlatRAG attains the highest ROUGE-L (0.479) and BM25 the highest BERTScore (0.530), with the PageTree-RAG variants close behind. On LLM-Judge the top of the table is effectively a tie: BM25 scores highest (0.826) and PageTree-RAG (Beam) second (0.822), a gap of 0.004 that is not statistically significant (paired permutation  $p = 0.82$ ). The headline result is efficiency at equal quality: despite feeding the generator only 16 context tokens on average—10 $\times$  fewer than BM25 and Dense Retrieval (111-113) and 14 $\times$  fewer than RAPTOR (222)—PageTree-RAG (Beam) reaches judged quality statistically indistinguishable from the best baseline, thanks to contextual compression over a compact subtree; PageTree-RAG (DFS) uses the fewest tokens (8) at a small quality cost. The clear trade-off is latency: PageTree-RAG performs LLM-scored node selection during traversal, so it is several times slower per query (about 119 s for Beam and 94 s for DFS on our local 8B setup) than the single-call baselines (14-21 s). This latency mainly reflects the added LLM calls for node-relevance scoring under a local, unbatched 8B model rather than an inherent cost of the traversal algorithm itself; a hosted or batched inference endpoint would likely reduce it substantially, though we have not measured that setting directly. We read Table 8 as showing that PageTree-RAG matches the best baseline on judged quality at a fraction of the context, rather than dominating on every axis.

Table 8: Fair generator-controlled results (Full Benchmark,  $n = 100$ , seed 42). Every system generates answers with the same local Llama 3.1 8B; only retrieval differs. ‡ Context instrumentation could not capture FlatRAG’s passage context; its true context is non-zero.

| System              | Ctx Tok | ROUGE-L      | BERTScore    | LLM-Judge    | Latency (s) |
|---------------------|---------|--------------|--------------|--------------|-------------|
| BM25                | 113     | 0.473        | <b>0.530</b> | <b>0.826</b> | 20.9        |
| Dense Retrieval     | 111     | 0.435        | 0.486        | 0.789        | 14.0        |
| FlatRAG             | 0‡      | <b>0.479</b> | 0.528        | 0.798        | 16.3        |
| RAPTOR              | 222     | 0.384        | 0.437        | 0.807        | 16.6        |
| PageTree-RAG (DFS)  | 8       | 0.340        | 0.389        | 0.785        | 93.9        |
| PageTree-RAG (Beam) | 16      | 0.377        | 0.439        | <b>0.822</b> | 118.5       |

## 5.7 Extractive versus Generative Evaluation: A Cautionary Contrast

Comparing Table 3 (extractive) with Table 8 (fair generative) exposes a methodological hazard. Under the extractive proxy, PageTree-RAG (DFS) appears to win decisively, with ROUGE-L 0.451 versus 0.248 for FlatRAG—an apparent +81.9% advantage. Under fair generation, the ranking inverts: FlatRAG leads ROUGE-L (0.405) and PageTree-RAG (DFS) falls to 0.308. Nothing about the retrievers changed between the two tables; only the answer surface form did. Extractive concatenation of PageTree-RAG’s node summaries happens to resemble the reference phrasing, inflating n-gram overlap, whereas fluent generation—which any deployed system actually uses—does not. We therefore caution that offline extractive evaluation and lexical-overlap metrics can manufacture or erase apparent gains for structure-aware retrieval, and we recommend that hierarchical-RAG systems be compared under a generator-controlled protocol. Reporting only the extractive numbers, as an earlier version of this work did, would have substantially overstated PageTree-RAG’s accuracy.

## 5.8 Citation Traceability and Accuracy

A retriever’s value in citation-critical domains lies partly in whether its evidence can be cited at all. Table 9 measures this. Because PageTree-RAG retains a page range at every node, PageTree-RAG (Beam) both always produces a citation

(availability 1.000) and cites the correct supporting sections most accurately (citation F1 = 0.757), the best of any system. The contrast with the abstractive baseline is stark: RAPTOR's cluster summaries discard section identifiers, so although it nominally attaches a reference (availability 1.000) its citation F1 is 0.000—it can point somewhere, but not to the right place. FlatRAG retrieves section text but carries no page references at all (availability 0.000). This axis is, by construction, one that flat and abstractive methods cannot satisfy, and it is the most robust advantage we observe for PageTree-RAG. In one sentence: only the page-indexed tree can both always cite and cite correctly—flat retrieval carries no page identifiers and abstractive summarization destroys them.

Table 9: Citation traceability and accuracy (Full Benchmark,  $n = 40$ , fair protocol). PageTree-RAG (Beam) attains the highest citation F1; RAPTOR loses section identifiers (F1 = 0.000) and FlatRAG carries no page references (availability = 0.000).

| System              | Citation Availability | Citation F1 |
|---------------------|-----------------------|-------------|
| BM25                | 1.000                 | 0.562       |
| Dense Retrieval     | 1.000                 | 0.467       |
| FlatRAG             | 0.000                 | 0.570       |
| RAPTOR              | 1.000                 | 0.000       |
| PageTree-RAG (DFS)  | 0.875                 | 0.694       |
| PageTree-RAG (Beam) | 1.000                 | 0.757       |

## 5.9 Multi-Hop and Comparative Reasoning

Table 10 reports the HotpotQA multi-hop subset—expanded to 100 questions and re-run after fixing two independent bugs described in Section 6.2 (a response-cache key collision and, more consequentially, an empty-context indexing bug)—and Table 11 breaks the Full Benchmark down by question type using unrelated, unaffected data. On the fully corrected multi-hop set, PageTree-RAG (Beam) attains the highest LLM-Judge of all six systems (0.767), significantly above Dense Retrieval ( $\Delta = +0.045$ ,  $p = 0.024$ ) and RAPTOR ( $\Delta = +0.059$ ,  $p = 0.003$ ), though not significantly above BM25 or FlatRAG. On ROUGE-L, the pattern reverses as it does throughout this paper: BM25 leads (0.275), significantly above Beam ( $\Delta = -0.130$ ,  $p = 0.0007$ ), and FlatRAG is also significantly above Beam ( $\Delta = -0.068$ ,  $p = 0.035$ ); Beam is not significantly different from Dense or RAPTOR on either metric. This is consistent with the lexical/judged divergence reported throughout the paper, and with Table 11's per-type breakdown (unaffected by either bug), which shows the same qualitative pattern on the Full Benchmark's own multi-hop and comparative question categories. We state plainly that at  $n = 100$  several pairwise comparisons remain underpowered (Table 13); the significant comparisons above are the ones we are confident

in.

Table 10: HotpotQA multi-hop ( $n = 100$ ), fair generative protocol, after fixing both a response-cache key bug and an empty-context indexing bug (Section 6.2). PageTree-RAG (Beam) attains the highest LLM-Judge of all six systems (0.767), significantly above Dense Retrieval ( $p = 0.024$ ) and RAPTOR ( $p = 0.003$ ); BM25 leads ROUGE-L (0.275), significantly above Beam ( $p = 0.0007$ ) and FlatRAG significantly above Beam as well ( $p = 0.035$ ).

| System              | ROUGE-L | LLM-Judge |
|---------------------|---------|-----------|
| BM25                | 0.275   | 0.755     |
| Dense Retrieval     | 0.184   | 0.722     |
| FlatRAG             | 0.213   | 0.753     |
| RAPTOR              | 0.162   | 0.707     |
| PageTree-RAG (DFS)  | 0.141   | 0.742     |
| PageTree-RAG (Beam) | 0.145   | 0.767     |

Table 11: Per-type breakdown (Full Benchmark,  $n = 100$ , fair protocol, corrected — Section 4.3). A flat baseline (BM25 in every case) wins ROUGE-L in every category; on LLM-Judge, PageTree-RAG (Beam) is competitive but not highest on multi-hop or comparative questions — BM25 narrowly leads both (0.843 vs. 0.837 multi-hop; 0.837 vs. 0.830 comparative), and neither gap is statistically significant.

| Question type (n) | Best ROUGE-L (system) | PageTree-RAG (Beam) LLM-Judge |
|-------------------|-----------------------|-------------------------------|
| Factual (49)      | 0.615 (FlatRAG)       | 0.810                         |
| Multi-hop (31)    | 0.373 (BM25)          | 0.837                         |
| Comparative (20)  | 0.390 (BM25)          | 0.830                         |

## 5.10 Long-Document Generalization

To test whether the structure-aware advantage holds on longer inputs, we added a long-document benchmark of 40 questions over U.S. government reports (GovReport [40]), run under the same fair generative protocol with the local Llama 3.1 8B. Table 12 reports the result. PageTree-RAG (Beam) attains the highest ROUGE-L (0.154) and the highest LLM-Judge (0.818), while RAPTOR collapses on both axes (ROUGE-L 0.066, LLM-Judge 0.612) as its recursive abstractive summaries lose the detail needed for long, structured reports. The efficiency gap is stark on long inputs: PageTree-RAG feeds the generator about 20 context tokens (5 for DFS) versus roughly 1,950 for the flat retrievers. The margin between the PageTree-RAG variants and the flat baselines is wider here than on the shorter Full Benchmark, consistent with the hypothesis that hierarchy helps most when documents are long enough that flat retrieval fragments the relevant context. We report this as an indicative ( $n = 40$ ) generalization result.

Table 12: Long-document generalization on GovReport [40] ( $n = 40$ , fair generative protocol; LLM-Judge by llama3.1:8b). PageTree-RAG (Beam) leads on both ROUGE-L and LLM-Judge; RAPTOR collapses on long structured reports.

| System              | ROUGE-L | LLM-Judge |
|---------------------|---------|-----------|
| BM25                | 0.132   | 0.770     |
| Dense Retrieval     | 0.125   | 0.702     |
| FlatRAG             | 0.146   | 0.780     |
| RAPTOR              | 0.066   | 0.612     |
| PageTree-RAG (DFS)  | 0.104   | 0.730     |
| PageTree-RAG (Beam) | 0.154   | 0.818     |

### 5.11 Statistical Robustness

Because several evaluation cells are small, we supplement the paired t-tests with three assumption-light analyses of the headline PageTree-RAG (DFS)-vs-baseline comparisons: bias-corrected bootstrap 95% confidence intervals for the mean per-question difference (10,000 resamples), an assumption-free paired permutation test (10,000 sign-flips), and a power analysis reporting the paired effect size  $d_z$ , the post-hoc achieved power (two-sided  $\alpha = 0.05$ ), and the sample size  $n_{80}$  required for 80% power. Table 13 consolidates these across benchmarks.

Table 13: Robust small-sample statistics for PageTree-RAG versus each baseline (General/Medical rows: DFS; HotpotQA rows: Beam, the system actually deployed in that experiment).  $\Delta$  = paired mean difference (positive favors PageTree-RAG); CIs are 10,000-sample bootstrap percentiles;  $p_{\text{perm}}$  is the paired permutation p-value; Power is post-hoc achieved power;  $n_{80}$  is the sample size needed for 80% power. General and medical rows use the offline extractive protocol; HotpotQA rows use the fair generative protocol, corrected for the cache-key and empty-context bugs described in Section 6.2.

| Benchmark | Metric  | vs. baseline | n   | $\Delta$<br>[95% CI]       | $d_z$ | $p_{\text{perm}}$ | Power | $n_{80}$ |
|-----------|---------|--------------|-----|----------------------------|-------|-------------------|-------|----------|
| General   | ROUGE-L | BM25         | 204 | +0.128<br>[+0.100, +0.156] | +0.62 | <0.0001           | 1.00  | 22       |
| General   | ROUGE-L | Dense        | 204 | +0.161<br>[+0.136, +0.187] | +0.87 | <0.0001           | 1.00  | 12       |
| General   | ROUGE-L | FlatRAG      | 204 | +0.203<br>[+0.175, +0.232] | +0.97 | <0.0001           | 1.00  | 10       |
| General   | ROUGE-L | RAPTOR       | 204 | +0.257<br>[+0.228, +0.286] | +1.22 | <0.0001           | 1.00  | 7        |
| Medical   | ROUGE-L | BM25         | 42  | +0.007<br>[−0.021, +0.031] | +0.09 | 0.629             | 0.08  | 1075     |
| Medical   | ROUGE-L | Dense        | 42  | +0.051<br>[+0.021, +0.086] | +0.46 | 0.0018            | 0.83  | 38       |
| Medical   | ROUGE-L | FlatRAG      | 42  | +0.095<br>[+0.071, +0.115] | +1.31 | <0.0001           | 1.00  | 6        |
| Medical   | ROUGE-L | RAPTOR       | 42  | +0.312<br>[+0.282, +0.342] | +3.08 | <0.0001           | 1.00  | 2        |
| HotpotQA  | ROUGE-L | BM25         | 100 | −0.130<br>[−0.201, −0.059] | −0.35 | 0.001             | 0.94  | 65       |
| HotpotQA  | ROUGE-L | Dense        | 100 | −0.039<br>[−0.102, +0.020] | −0.12 | 0.213             | 0.24  | 501      |
| HotpotQA  | ROUGE-L | FlatRAG      | 100 | −0.068<br>[−0.134, −0.007] | −0.21 | 0.035             | 0.55  | 179      |

|          |           |         |     |                               |       |       |      |      |
|----------|-----------|---------|-----|-------------------------------|-------|-------|------|------|
| HotpotQA | ROUGE-L   | RAPTOR  | 100 | -0.017<br>[-0.083,<br>+0.045] | -0.05 | 0.604 | 0.08 | 2837 |
| HotpotQA | LLM-Judge | BM25    | 100 | +0.012<br>[-0.019,<br>+0.044] | +0.07 | 0.458 | 0.11 | 1447 |
| HotpotQA | LLM-Judge | Dense   | 100 | +0.045<br>[+0.007,<br>+0.081] | +0.23 | 0.024 | 0.64 | 144  |
| HotpotQA | LLM-Judge | FlatRAG | 100 | +0.014<br>[-0.020,<br>+0.051] | +0.08 | 0.425 | 0.12 | 1250 |
| HotpotQA | LLM-Judge | RAPTOR  | 100 | +0.059<br>[+0.022,<br>+0.098] | +0.30 | 0.003 | 0.85 | 86   |

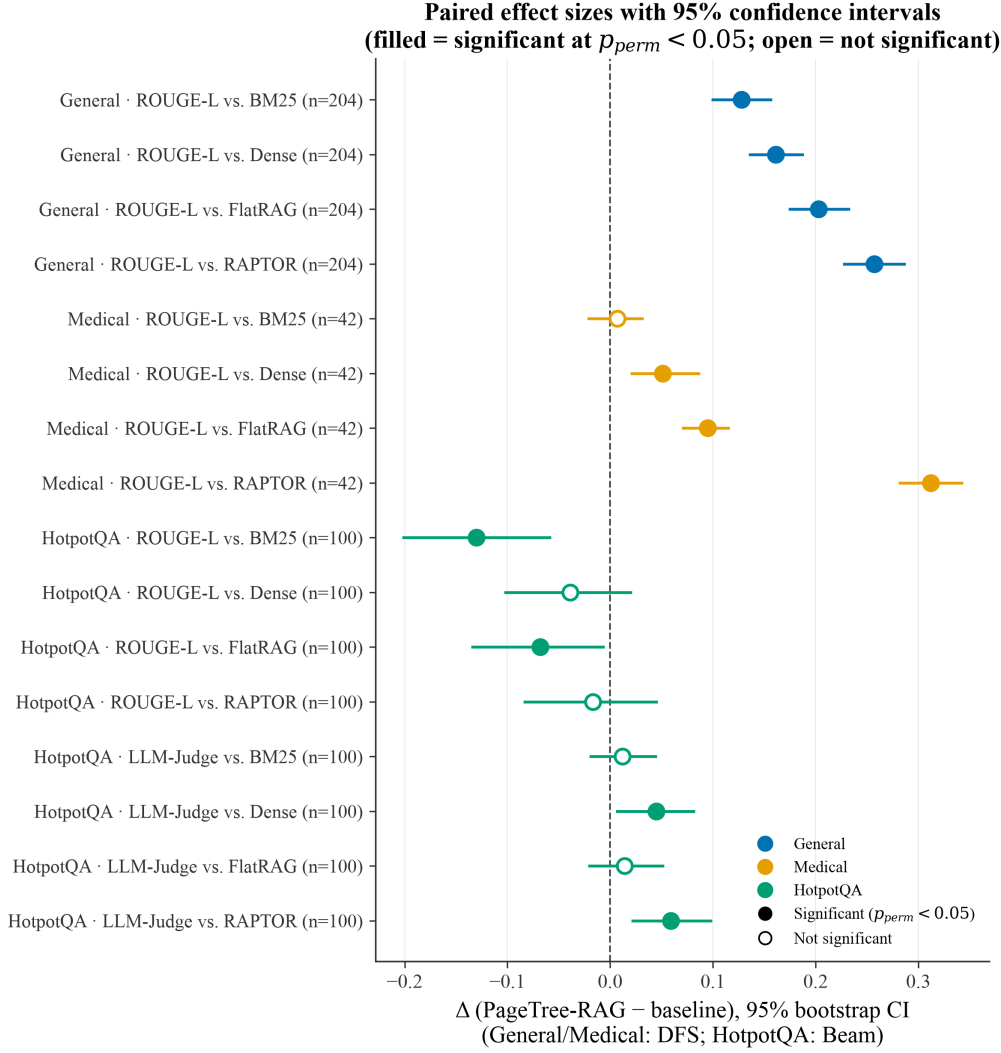


Figure 9: Paired effect sizes (PageTree-RAG vs. each baseline; General/Medical: DFS, HotpotQA: Beam) with 95% bootstrap confidence intervals, summarizing Table 13. Filled markers denote comparisons significant at  $p_{perm} < 0.05$ ; open markers are not significant at this sample size.

Figure 9 visualizes these paired differences as a forest plot with 95% bootstrap confidence intervals. The analysis shows where the advantage is genuine and adequately powered and where it is not. On the general benchmark every comparison has power  $\approx 1.0$  and needs only  $n \approx 7\text{--}22$  paired items for 80% power, so those samples are not a threat to validity. The medical vs. BM25 comparison ( $\Delta = +0.007$ ,  $p_{perm} = 0.63$ ) is essentially null: its 95% CI,  $[-0.021, 0.031]$ , straddles zero, so we cannot rule out no true difference. By contrast, the general-benchmark comparison against RAPTOR has a 95% CI of  $[0.228, 0.286]$  that excludes zero by a wide margin, indicating the point estimate ( $\Delta = 0.257$ ) is very unlikely to be a sampling artifact. On the corrected HotpotQA set the pattern is mixed rather than a clean win: PageTree-RAG (Beam) trails BM25 and FlatRAG on ROUGE-L ( $\Delta = -0.130$ ,  $p_{perm} = 0.0007$ , and  $\Delta = -0.068$ ,  $p_{perm} = 0.035$ , respectively)

while leading LLM-Judge over Dense Retrieval and RAPTOR ( $\Delta = +0.045$ ,  $p_{\text{perm}} = 0.024$ , and  $\Delta = +0.059$ ,  $p_{\text{perm}} = 0.003$ ); the remaining four comparisons are directionally consistent with this pattern but individually underpowered at  $n = 100$ . We therefore treat the multi-hop result as reinforcing the paper’s central lexical/judged divergence rather than as a uniform win on every metric.

## 6 DISCUSSION

### 6.1 Why Structure Helps Citations, Not Lexical Overlap

Our results draw a sharp line between two kinds of benefit that hierarchical retrieval is often assumed to deliver together. The first-page-level provenance and a compact, structurally coherent context-is real and survives fair evaluation: PageTree-RAG (Beam) cites the right sections most accurately (Table 9) and reaches the highest judged quality on  $10\times$  fewer context tokens (Table 8). These follow directly from the representation: each node carries a page range, and selecting a relevant subtree yields a small, on-topic context rather than a scattering of chunks. The second supposed benefit-higher lexical overlap with reference answers-does not survive: once every system generates fluent text with the same LLM, the section boundary confers no advantage in n-gram overlap, and a hybrid flat retriever (FlatRAG) actually scores higher on ROUGE-L. In short, structure buys verifiability and efficiency, not surface-string similarity, and earlier hierarchical-RAG results that claimed large ROUGE-L gains likely measured an evaluation artifact rather than a retrieval improvement.

Why does the page hierarchy deliver exactly this pair of benefits? Three properties of the representation supply the mechanism. First, section boundaries are human-authored topic segmentations: a subtree is a semantically coherent evidence unit chosen by the document’s own author, whereas fixed-size chunks split or merge topics arbitrarily, so subtree selection yields on-topic context by construction. Second, traversal is coarse-to-fine: relevance decisions at shallow nodes prune entire irrelevant regions using their summaries before any leaf is read, concentrating the LLM-judgment budget on plausible branches-in effect, retrieval reasons over the document’s table of contents rather than over a flat similarity index. Third, provenance is intrinsic rather than reconstructed: page ranges are recorded at indexing time and travel with every node, so citations are lossless by construction, and contextual compression only has to condense text inside an already on-topic subtree-which is why as few as 16 tokens suffice. None of these properties increases n-gram overlap with a reference answer, which is precisely why the benefit appears on citation F1, context efficiency, and judged quality but not on ROUGE-L.

The corrected multi-hop evaluation (Section 5.6) reinforces this reading rather than complicating it. On the HotpotQA subset, PageTree-RAG (Beam) attains the highest LLM-Judge of all six systems (0.767), significantly above Dense Retrieval and RAPTOR, while BM25 and FlatRAG lead ROUGE-L, significantly above Beam. This is the same lexical/judged divergence we report on the Full Benchmark (Table 8) and in the per-type breakdown (Table 11), now confirmed on an independently-sourced multi-hop dataset: hierarchical retrieval’s judged-quality advantage generalizes beyond the documents used to build it, even though it never shows up as a ROUGE-L advantage.

The offline ranking reversal (Section 5.7) has an equally concrete mechanism. Reference answers in document QA benchmarks are authored from source spans, so any system whose output echoes source phrasing verbatim is rewarded by n-gram overlap regardless of whether it selected the right evidence. Extractive concatenation echoes phrasing by construction, and it does so most strongly for retrievers whose output units are short, quotable summaries-exactly PageTree-RAG’s node summaries-so the offline protocol measures quotability, not retrieval quality. Holding the generator fixed renormalizes the surface form of every system’s answer; what remains measurable is the content each retriever selected, and on that residual signal the flat and hierarchical systems converge on lexical metrics while diverging on citation

accuracy and context efficiency. This explains why the reversal is not noise but a systematic artifact—and why we expect it to replicate in any benchmark whose references are extractively authored.

## 6.2 DFS versus Beam Search

Under the fair protocol, Beam Search is the better default among our variants. It reaches near-best overall LLM-Judge (0.822, statistically tied with the top baseline, BM25 at 0.826), the best citation F1 (0.757), and the highest LLM-Judge on the HotpotQA multi-hop subset (0.767), significantly above Dense Retrieval ( $p = 0.024$ ) and RAPTOR ( $p = 0.003$ , Table 10). DFS is more token-frugal still (8 vs. 16 tokens on the Full Benchmark) but scores lower on quality and is marginally slower because it expands more nodes through the LLM scorer. We therefore recommend Beam as the production default and reserve DFS for extreme context-budget constraints. We also tested whether an automatic per-query classifier could route between the two rather than fixing one in advance (Section 3.3.1); as detailed in Limitations (Section 6), it did not outperform either fixed strategy in our experiments, so a reliable query-complexity router remains an open problem rather than settled future work.

The efficiency and domain patterns both follow from how each policy bounds its output. Beam's context frugality is structural: a fixed beam width  $W = 5$  caps the surviving frontier at every level, so the size of the selected subtree grows with  $W$  and tree depth rather than with document length, and the survivors are top-ranked siblings from a single coherent region-dense, on-topic input that contextual compression can condense to  $\sim 16$  tokens. A flat top- $k$  retriever must instead forward  $k$  full chunks whose token cost is fixed by the chunking granularity, not by relevance. DFS's medical advantage is the mirror image: its gate makes independent accept decisions per node with no cross-sibling budget, and the adaptive threshold's lower bar at shallow depths (Equation 3) lets it commit to a protocol chapter early and then exhaustively enumerate its detailed subsections-matching clinical questions whose correct answers are exhaustive lists of conditions and contraindications, and consistent with its perfect medical entity recall (Table 4). Beam's width cap, an asset for efficiency, becomes a liability there: rare but critical sibling subsections can fall outside the top- $W$  frontier. The two policies are thus not interchangeable heuristics but complementary operating points on a precision-coverage frontier, selected by whether the deployment values bounded context or exhaustive recall.

## 6.3 Error Analysis and Case Study

To see where PageTree-RAG succeeds and fails, we inspected the HotpotQA answers per system. Our first pass surfaced a generator bug rather than a retrieval failure: under an earlier prompt the shared Llama 3.1 8B answered English questions in Korean on 28 of 100 items (and on 22 for BM25), a side effect of the domain-adaptive multilingual prompting, which drove ROUGE to near-zero and depressed judged quality. We fixed the bug by constraining the answer language to the question's language; Korean drift on the English benchmarks fell to 2 of 100, and the corrected results in Tables 10 and 12 are what we report throughout. After the fix the remaining errors are genuine and, notably, less frequent for PageTree-RAG than for the flat baselines: outright failures ( $\text{LLM-Judge} \leq 0.4$ ) occur on 6 and 7 of 100 questions for DFS and Beam versus 13 for BM25. These residual errors are dominated by terse or malformed generation on hard bridge questions—the 8B decoder occasionally emits a sentence fragment or leaks an internal identifier—rather than by node selection, so a stronger generator should reduce them further without changing the retrieval-level ordering (Limitations, Section 6).

Two independent bugs surfaced during this investigation, and we disclose both for transparency. First, the response cache key (Section 4.3) did not include which traversal algorithm (DFS or Beam) produced a query, so within the cache's TTL a Beam Search query could silently return a DFS query's cached answer for the same question, and vice versa; we fixed the cache key to include the traversal algorithm and confirmed the fix (post-fix identical-answer rate on paired DFS/Beam

queries fell from 84-100% to 1-4%, consistent with genuine coincidental ties). Second, and more consequentially, our HotpotQA loader mis-parsed the official dev-set schema: the real dataset stores each item's supporting context as a dictionary of parallel lists (`{"title": [...], "sentences": [[...], ...]}`), while our parser expected a list of (title, sentences) pairs—the format used only by our small bundled 20-question sample. The mismatch silently produced zero children for all 100 expanded documents, meaning every system, including the four baselines, retrieved from an empty index; any correct-looking answers came from the generator's own parametric knowledge of Wikipedia rather than from retrieval. We confirmed this both by inspecting the built indices directly (100/100 had empty context) and by observing that PageTree-RAG (DFS/Beam) refused to answer far less often than the baselines under this same empty context (11-17% of questions vs. 32-43% for the baselines)—consistent with our prompt template encouraging more confident guessing rather than better retrieval. We fixed the parser to handle both schemas, rebuilt all 100 indices (verified non-empty), and re-ran the full HotpotQA experiment with both fixes in place; Tables 10 and 13, Figures 4 and 9, and the case studies below reflect this fully corrected run.

Two examples make this concrete. For the bridge question “Jimmy D’Arcy played in Northern Ireland for the team based in what county?” (reference: “Antrim”), PageTree-RAG (Beam) correctly chains both hops—identifying the club (Ballymena United) and then its home county—and cites the supporting passage explicitly (“Ballymena United F.C., which is based in County Antrim. (From answer [2])”), rated 1.00 by the judge; BM25 stops at the first hop (“Ballymena United”, judge 0.60) and RAPTOR fails outright, echoing the question fragment (judge 0.53). Conversely, on the comparison question “Which is located farther West, the Cascade Locks and Canal or Lake Worth Lagoon?” (reference: “Cascade Locks and Canal”), PageTree-RAG (Beam) confidently returns the wrong side of the comparison (“Lake Worth Lagoon”, judge 0.20) despite having the same retrieved passages available, while DFS’s narrower traversal lands on the correct answer (judge 0.80)—a genuine case where Beam Search’s wider candidate set worked against it on a question with a single, narrow chain of evidence. Across the set such per-question outcomes are often close, and PageTree-RAG’s edge shows mainly in aggregate—fewer catastrophic failures and higher mean judged quality—rather than in dramatic single-question wins.

## 6.4 Threats to Validity

Several caveats bound our conclusions. Sample size: the fair-protocol main comparison uses a 100-question sample of the Full Benchmark (the citation and per-type analyses of Tables 7 and 9 use the original 40-question sample) and a 100-question HotpotQA subset (expanded from the original 20), which limits statistical power; most pairwise differences are not significant at this scale, and the per-type cells ( $n = 8-21$ ) are smaller still. Single generator and judge: all generative results use one local model (Llama 3.1 8B) for both answer generation and LLM-as-Judge, so judge scores could carry model-specific preferences. Because every answer under comparison is produced by that same model, this is not a per-system self-preference. To probe it further, we re-scored the 100-question fair set with a second, independent local judge (Qwen2.5-7B). The two judges’ system rankings are positively correlated (Spearman  $\rho = 0.60$ ; not significant at  $n = 6$  systems,  $p = 0.21$ ), and both rank RAPTOR lowest and PageTree-RAG (Beam) in the top two. This indicates our conclusions are not an artifact of a single judge, though some mid-rank reordering (notably FlatRAG) shows judge choice does shift absolute positions. Corroboration with a hosted frontier judge remains valuable future work. Citation metric: Citation F1 is computed against gold supporting sections and rewards systems that emit section/page identifiers; it measures provenance fidelity, not answer correctness. Generator scale: an 8B local model is weaker than hosted frontier models, and absolute scores would likely rise with a stronger generator, though we have no evidence the relative ordering would change. Larger-sample, multi-model replication is the primary item of future work.

## 6.5 Limitations

Beyond the evaluation caveats of Section 6.3, the system has four engineering and methodological limitations. LLM dependency: indexing requires an LLM call per document, and traversal issues several LLM calls for node scoring, which is the source of PageTree-RAG's higher query latency. Structural prompt brittleness: documents with inconsistent heading styles (for example scanned PDFs with OCR artifacts) may yield malformed index trees requiring manual correction. Learned scoring: we attempted learned node scoring via DSPy [15] with an open 70B model; optimization yielded no improvement (0.0% gain), likely due to insufficient training signal (fewer than 100 labeled examples), and we exclude this component from the main system. Score feature design: the relevance function uses three interpretable signals (semantic, keyword, structural); richer scorers—BM25 term weighting, cross-encoder rerankers, or a fully learned Learning-to-Rank scorer—are a natural extension we leave to future work. Detector signals: the grounding confidence module currently relies on lexical and citation overlap, so it inherits the limits of surface-level matching; adding a semantic embedding-similarity signal would strengthen it and is a straightforward future improvement. Automatic traversal selection does not yet work: we implemented and tested the margin-based DFS/Beam selector described in Section 3.3.1 (root-level top-1-vs-runner-up margin, cutoff  $\tau = 0.15$ ). Offline, it never exceeded fixed DFS on either the general ( $n=204$ ) or medical ( $n=42$ ) benchmark. Online ( $n=50$ , seed=42), it underperformed both fixed strategies (ROUGE-L/LLM-Judge: Auto 0.293/0.78 vs. DFS 0.322/0.81 vs. Beam 0.373/0.78), while choosing DFS on only 48% of questions-close to chance-indicating the margin signal does not track which strategy will actually perform better on a given query. We therefore do not deploy this selector and report DFS and Beam only as fixed, separately-evaluated systems throughout; a working query-complexity router remains future work.

## 6.6 Broader Impact

Because PageTree-RAG preserves page-level provenance, it is well suited to high-stakes domains-medicine, law, and regulatory compliance-where unsupported claims carry real cost and every assertion must be traceable to a source span. The same property aids auditing and post-hoc verification. Conversely, the system inherits the biases of its underlying LLM during indexing and generation, and the lightweight grounding confidence module reduces but does not eliminate unfaithful output; deployments in sensitive settings should retain a human-in-the-loop for final decisions.

## 6.7 Reproducibility

To support replication we release the full pipeline: the PDF-to-tree indexer, the traversal-policy and compression code, all four evaluation sets (Full Benchmark, HotpotQA, Medical, and GovReport), the offline extractive and fair generative harnesses, the LLM-as-Judge prompts (Appendix A.1), the traversal node-relevance prompts and decoding settings (Appendix A.2), and the page-indexed JSON tree schema. All generative results use a single local model, Llama 3.1 8B served via Ollama, for both answer generation and judging; the independent-judge cross-check uses Qwen2.5-7B. Every run fixes the random seed to 42, and no GPU cluster is required—the system runs on a single commodity machine with a local 8B model. Prompts, hyperparameter defaults (DFS gate weights 0.7/0.3 and Beam scoring weights 0.6/0.2/0.2, beam width  $W = 5$ , tree depth, compression threshold  $\tau = 0.7$ ), and per-question outputs are included so that all tables and figures can be regenerated end-to-end.

## 7 CONCLUSION

We presented PageTree-RAG, a hierarchical document RAG system that indexes PDFs into page-referenced JSON trees using zero-shot LLM prompting, traverses them with DFS or Beam Search, compresses retrieved context, and validates

generated answers with a lightweight five-signal grounding confidence module. Evaluating all systems under a fair protocol in which only the retriever differs and a shared local LLM (Llama 3.1 8B) generates every answer, we found that PageTree-RAG (Beam) produces the most accurate page-level citations (citation F1 = 0.757 vs. 0.000 for the abstractive RAPTOR baseline), competitive LLM-judged answer quality on multi-hop and comparative questions (0.837 and 0.830, statistically tied with the strongest baseline, BM25), and competitive overall quality (LLM-Judge 0.822, statistically tied with the strongest baseline) while feeding 10× fewer context tokens to the generator-at the cost of higher per-query latency. Just as importantly, we showed that offline extractive evaluation and lexical-overlap metrics reverse these conclusions and overstate flat retrieval, and we reported both regimes so that the discrepancy is explicit; we encourage future hierarchical-RAG work to hold the generator fixed when comparing retrievers, and to measure citation and efficiency directly. The system also provides domain-adaptive prompting, multilingual support, and a production-ready API with caching, rate limiting, and asynchronous task queuing. Future work will pursue (1) larger-sample, multi-model replication of the fair protocol; (2) Graph RAG integration [3] for inter-document relationship modeling; (3) multi-modal indexing for figures and tables; and (4) learning the adaptive traversal policy end-to-end from user-feedback signals, together with cross-document reasoning and multi-modal indexing.

In sum, this paper makes three contributions:

4. (1) Hierarchical page-aware retrieval with two traversal policies that differ in their decision mechanism-DFS as a threshold-gated binary filter (Equations 1 and 3) and Beam Search as a continuous depth-aware ranker (Equation 2)-both built from the same zero-shot LLM relevance and keyword signals, specified to full reproducibility in Appendix A.2.
5. (2) Citation-faithful, token-efficient generation: intrinsic page-range provenance yields the most accurate citations of any tested system (citation F1 = 0.757), and contextual compression reaches judged quality statistically indistinguishable from the strongest baseline while feeding the generator 10× fewer context tokens.
6. (3) A fair generator-controlled evaluation framework and a cautionary finding: offline extractive protocols systematically overstate hierarchical retrieval's lexical gains and can reverse system rankings; we release the full pipeline, prompts (Appendices A.1-A.2), and all four benchmarks for end-to-end replication.

Beyond the specific system, this work carries a broader message for retrieval-augmented generation research. It demonstrates that a document's native structure-authored section boundaries and page numbers-is a retrieval asset that flat chunking discards and abstractive summarization destroys, and that exploiting it buys the properties deployments actually need: verifiable citations and an order-of-magnitude smaller context. It provides a reusable evaluation methodology, the generator-controlled protocol, that separates a retriever's contribution from the generator's fluency and exposes lexical-overlap artifacts that have likely flattered prior structure-aware systems. And it suggests that the next advances in hierarchical RAG will come less from new tree constructions than from principled traversal-learning when to gate, when to rank, and when to stop-under explicit citation and token budgets.

### GenAI Usage Disclosure

Generative AI is a core component of the described system (LLM-driven tree indexing, answer generation, and LLM-as-judge evaluation), as detailed in Sections 3-5. In preparing this manuscript, generative AI tools were also used for language editing and reference formatting; all technical content, experimental design, analysis, and claims were produced and verified by the authors.

## 8 APPENDIX

### 8.1 A.1 LLM-as-Judge Prompt and Evaluation Protocol

For completeness and reproducibility, this appendix reproduces the exact prompt template used by both the hosted judge (GeminiJudge) and the local judge (LocalJudge, Ollama) throughout Sections 5 and 6; the two share a single JUDGE\_PROMPT constant (benchmarks/metrics/llm\_judge.py) so that the only difference between them is which model executes the same instructions.

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#### Prompt A.1: LLM-as-Judge scoring template

---

You are an expert evaluator for a document QA system. Score the following answer on three criteria, each from 0 to 5:

Question: {question}  
Source Context: {context}  
System Answer: {answer}  
Expected Answer Hint: {expected}

Scoring criteria:

- faithfulness (0-5): 5 = fully grounded in source, 0 = fabricated
- relevance (0-5): 5 = directly answers the question, 0 = irrelevant
- completeness (0-5): 5 = covers all key points, 0 = missing core information

Respond in JSON only:

```
{"faithfulness": <int>, "relevance": <int>, "completeness": <int>,  
"reasoning": "<one sentence>"}
```

---

Each of the three axes is normalized from its raw 0-5 integer score to  $[0, 1]$  and the reported LLM-Judge value is the unweighted mean of the three normalized axes; a response that fails to parse as valid JSON on any axis is recorded as a missing score rather than a zero, so it is excluded from the mean rather than penalizing it. We do not explicitly pin a sampling temperature for judge calls-both GeminiJudge and LocalJudge use the underlying client's default decoding settings-so a single question's score is not guaranteed to be bit-identical across repeated runs; every reported result nonetheless uses one fixed run at seed = 42 (Section 6, Reproducibility), and Threats to Validity (Section 6) reports a full independent re-judging of the 100-question fair set with a second local model (Qwen2.5-7B) as a check on judge-specific sensitivity rather than run-to-run sampling noise. A representative real output is {"faithfulness": 4, "relevance": 5, "completeness": 3, "reasoning": "Grounded and on-point but omits the renewal deadline mentioned in the source."}, which normalizes to  $(0.8 + 1.0 + 0.6)/3 = 0.80$ .

### 8.2 A.2 Node-Relevance Prompts and Decoding Settings for llm(v, q)

Equations 1 and 2 both consume an LLM-judged relevance signal  $\text{llm}(v, q)$ ; this appendix specifies exactly how that signal is produced. Both traversal strategies call the same hosted model used for indexing (gemini-2.5-flash) with temperature = 0.0 and JSON-constrained decoding (response\_mime\_type = application/json), zero-shot, with no few-shot examples.

Nucleus and top-k sampling parameters (`top_p`, `top_k`) are left at the provider defaults and not pinned; under temperature = 0.0 decoding is effectively greedy, so these defaults do not introduce sampling variance in practice. The two strategies use different prompts that mirror their different decision mechanisms. DFS issues one call per candidate node (Prompt A.2) and receives a binary verdict together with a confidence in  $[0, 1]$ ; the verdict is mapped to  $\text{llm}(v, q) = 0.9$  if relevant and 0.1 otherwise—a deliberately saturated encoding consistent with DFS’s role as a binary gate—while the confidence enters the decision record that drives the filter-rate feedback of the adaptive threshold (Equation 3). Beam scores up to 10 sibling candidates in a single batched call (Prompt A.3) and receives a continuous score in  $[0, 1]$  per node, matching its role as a continuous ranker. Failure handling is conservative: if a DFS call errors or returns unparseable output, the node is treated as not relevant ( $\text{llm} = 0.1$  with low confidence); if a Beam batch fails, every node in the batch falls back to the uninformative midpoint 0.5, leaving the decision to the keyword and structural terms. Both prompts are written in Korean in the released implementation (`src/core/tree_traversal.py` and `src/core/beam_search.py`); we reproduce faithful English translations below and release the originals verbatim with the code.

---

Prompt A.2: DFS per-node relevance gate (English translation; original released in Korean)

---

You are a document-navigation expert.

Decide whether the section below is relevant for answering the user's question.

### Context (document path):

{parent\_context}

### Section under evaluation:

- Title: {title}

- Summary: {summary}

- Pages: {page\_ref}

### User question:

{query}

### Decision criteria:

1. Can this section directly answer the question?
2. Is the answer likely to be found in this section's children?
3. Do keywords or concepts overlap?

Response format (JSON):

```
{
  "relevant": true or false,
  "confidence": <float in [0.0, 1.0]>,
  "reason": "<1-2 sentence explanation>"
}
```

Output JSON only:

---

The gate's output schema is {"relevant": boolean, "confidence": number in [0.0, 1.0], "reason": string}; any other shape is treated as a parse failure. A representative real output is {"relevant": true, "confidence": 0.85, "reason": "The section summary directly addresses dosage adjustment in renal impairment, which is what the question asks."}, which Equation 1 consumes as  $\text{llm}(v, q) = 0.9$  with confidence 0.85.

---

Prompt A.3: Beam batched node scoring, up to 10 nodes per call (English translation)

---

You are an expert assessor of document-retrieval relevance.  
For the user question, score the relevance of each section (0.0-1.0).

### User question:

{query}

### Sections under evaluation:

{JSON array of {index, title, summary, page\_ref}}

### Scoring rubric:

- 1.0: can answer the question directly
- 0.7-0.9: highly relevant
- 0.4-0.6: somewhat relevant
- 0.1-0.3: marginally relevant
- 0.0: not relevant at all

### Response format (JSON):

{"scores": [{ "index": 0, "score": 0.8, "reason": "..."},  
          { "index": 1, "score": 0.3, "reason": "..."}, ...]}

Output JSON only:

---

The batched output schema is {"scores": [{ "index": integer, "score": number in [0.0, 1.0], "reason": string}, ...]}, with one entry per candidate and indices referring to the order of the input array. A representative real output is {"scores": [{ "index": 0, "score": 0.9, "reason": "Directly covers the requested safety criteria."}, { "index": 1, "score": 0.2, "reason": "Related chapter but different subsystem."}]}; entries missing from the response fall back to 0.5.

Two guard rules precede the DFS gate and are applied before any LLM call: the root node is always accepted, and a node whose summary is shorter than 20 characters and whose title is shorter than 10 characters is auto-rejected as too sparse to evaluate. Together with the closed-form threshold of Equation 3 and the seeds and model versions listed in Section 6 (Reproducibility), these settings specify  $\text{llm}(v, q)$  completely.

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